Optimisation, Fabrication, and Characterisation of AlGaN/GaN HEMTs

James Svanström
ja8310sv-s@student.lu.se

Department of Electrical and Information Technology
Lund University

Supervisor: Avinas N. Shaji

Examiner: Erik Lind

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Abstract

This thesis investigates the process optimisation, fabrication, and electrical characterisation of AlGaN/GaN high-electron mobility transistors (HEMTs). Before device fabrication, etching processes for AlGaN recess structures were optimised, followed by an analysis of SiN_x and Al_2O_3 for use in device isolation and in the gate stack. The fabricated devices incorporated intentional variations in their geometry. The influence of these variations was studied during the electrical characterisation of the devices.

The fabricated HEMTs demonstrated functional transistor operation with on-current and transconductance values of approximately $I_{on} \sim 300$ mA/mm and $g_m \sim 100$ mS/mm at $V_D = 3$ V. Increased source and drain recess depths were observed to reduce on-resistance, while increased gate recess depths showed to shift the devices towards normally-off behaviour, consistent with trends reported in the literature.

The devices exhibited low breakdown voltages and high leakage currents, which were attributed to limitations in the SiN_x passivation quality and a significant buffer conductivity. Field plate structures showed limited improvement in breakdown performance, indicating that further optimisation of their fabrication and integration may be required.

Overall, this work demonstrates the successful fabrication and characterisation of AlGaN/GaN HEMTs while identifying several process limitations affecting leakage, breakdown behaviour, and device variability. The results guide future process optimisation of GaN-based transistor technologies.

Popular Science Summary

Modern society increasingly relies on electrical energy. From smartphones to electric vehicles, countless technologies depend on electrical power conversion systems to operate. Electrification can help reduce carbon emissions when electricity is generated from low-carbon energy sources such as wind, solar, hydroelectric, or nuclear power.

Improving the efficiency of the power electronics used to convert and control electricity can reduce energy losses and lower operating costs. Gallium nitride (GaN) is a semiconductor material that can withstand high electric fields and enable more efficient power electronic devices compared to conventional devices fabricated from silicon. This makes it a promising candidate for the next generation of power transistors and diodes.

Despite their excellent performance, GaN-based devices remain challenging to manufacture reliably. Small variations introduced during fabrication can lead to significant differences in device performance, limiting their suitability for large-scale production.

This thesis investigates the fabrication of lateral high-electron-mobility transistors (HEMTs) based on an aluminium gallium nitride/gallium nitride (AlGaN/GaN) material stack. The devices were fabricated at the cleanroom facilities of Lund Nano Lab and demonstrated strong electrical performance. To improve manufacturing consistency, key fabrication processes were optimised before device fabrication. Electrical measurements of the fabricated devices revealed how device processing and transistor geometry influence performance. These findings provide valuable insight into improving fabrication processes and support the development of more reliable AlGaN/GaN power electronics.

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Furthermore, I would like to thank Myfab and NanoLund for providing access to the Lund Nano Lab facilities, and the laboratory staff for their assistance and training in the use of the cleanroom equipment.

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Chapter 1
Introduction

Decarbonisation has been identified by the European Union as a key objective for achieving sustainable industrial development [1]. One of the primary strategies for reducing greenhouse gas emissions is the electrification of transportation, energy generation, and industrial processes [2]. This ongoing transition places demands on power electronic systems, which must handle higher voltages, currents, and power densities while minimizing energy losses.

The performance of power electronic devices is limited by the properties of the semiconductor material from which they are fabricated [3]. Figure 1.1a compares material parameters relevant to power electronics for several established and emerging semiconductor materials, normalized to silicon. Figure 1.1b illustrates the corresponding theoretical limits of device performance imposed by material properties. The parameters used for these comparisons are described in Chapter 2 below.

Silicon (Si) has long been the dominant semiconductor material for power electronics due to its mature manufacturing ecosystem, low cost, and high reliability. However, as operating voltages and power densities continue to increase, silicon devices approach their intrinsic material limits. Silicon carbide (SiC), a wide-bandgap semiconductor, has enabled significant improvements in high-power applications and has become an established technology for medium- and high-voltage systems [7, 8]. Despite these advances, the trade-off between breakdown voltage and on-state resistance seen in Figure 1.1b remains a limiting factor for further improvements in efficiency and power density.

Gallium nitride (GaN) has therefore emerged as one of the most promising semiconductor materials for next-generation power electronics. Due to its wide band gap, high critical electric field, and high electron saturation velocity, GaN enables devices capable of operating at higher voltages and switching frequencies while maintaining conduction losses lower than comparable silicon-based technologies [9]. These properties make GaN particularly attractive for applications where efficiency and high-power operation are critical.

Among GaN-based device architectures, the AlGaN/GaN high-electron-mobility transistor (HEMT) has attracted attention. The heterostructure formed between AlGaN and GaN creates a high-density two-dimensional electron gas (2DEG) through spontaneous and piezoelectric polarization effects, resulting in the formation of a highly conductive channel without the need for doping [10]. AlGaN/GaN HEMTs thus offer the potential for high-performance power switching devices.

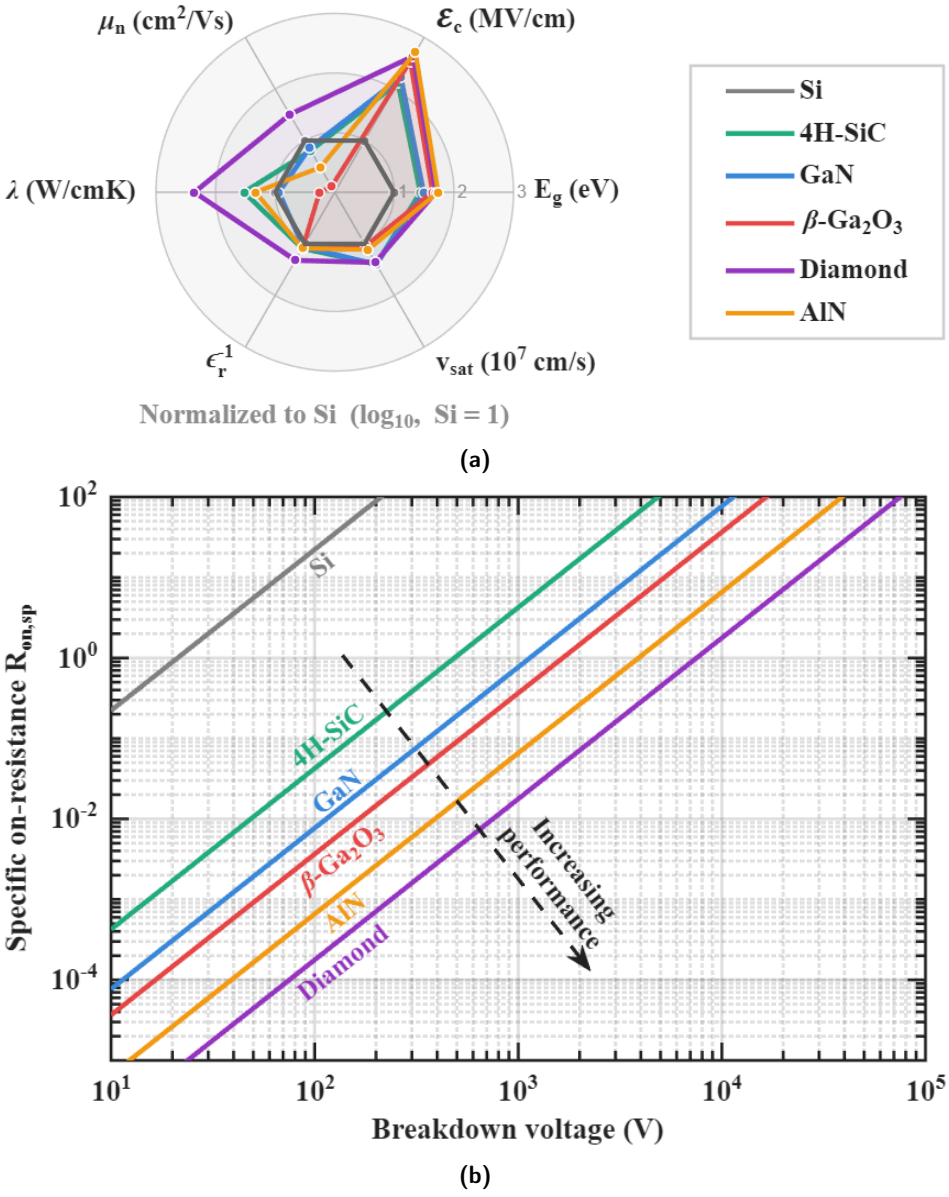


Figure 1.1: Two plots showing a) the material properties of common power device materials, normalised to silicon, and b) A constant Baliga Figure of Merit (BFOM) plot illustrating the theoretical trade-off between breakdown voltage and specific on-resistance, highlighting the performance limits of different power semiconductor materials. The displayed material properties are gathered from [4–6].

Despite their significant potential, GaN-based technologies have not yet reached the same level of industrial maturity as silicon and SiC. Regarding AlGaN/GaN HEMTs, device performance is highly dependent on the quality of the epitaxial heterostructure, where layer thicknesses, strain conditions, interface quality, and defect densities strongly influence the properties of the 2DEG and the resulting device characteristics [9]. Furthermore, challenges related to fabrication, gate dielectric integration, charge trapping, threshold voltage control, and long-term reliability continue to limit widespread adoption. As a result, both material development and process optimization remain active areas of research.

This thesis investigates lateral AlGaN/GaN HEMTs fabricated on an AlGaN/AlN/GaN heterostructure. The work focuses on the development and optimisation of the fabrication process, the realisation of standard Schottky-barrier HEMTs and metal-insulator-semiconductor HEMT (MIS-HEMT) devices. After fabrication, electrical characterisation is performed on the resulting structures, and the results are presented and discussed. The device processing was performed in the ISO 5 and ISO 7 clean room facilities at the Lund Nano Lab. During the course of the project, material-related challenges within the provided material stack were identified and combated. The characterised devices provided additional insight into the interplay between device fabrication, design philosophy, and electrical performance.

Chapter 2
Background

2.1 Field-effect transistors

Transistors are to the electrical circuit what a valve is to the mechanical pipeline: a way of controlling the flow of electrical current. Transistors are three-terminal devices that are fully electrically controlled, where the allowed current between two terminals of the device is modulated by the third terminal.

There are different species of transistors, differentiated by the mechanism of electrical control used to modulate the current travelling through the device. For this text, the field-effect transistor (FET) will be highlighted, specifically the high-electron mobility transistor (HEMT), which relies on field-effect electrical control.

An FET operates by applying a voltage between the third control terminal and the remaining terminals. The voltage creates an electrical field that then modulates the conductivity of the device. Typically, in FETs, the control terminal is referred to as the "gate", while the current flows between the "source" and "drain". A channel running under the gate contact connects the source and drain. The gate voltage modulates the channel charge population. When the channel is populated with a high concentration of charge carriers, the transistor is said to be in the on-state, allowing free flow of current between the source and drain through the channel [11]. Channels with a low concentration of charge carriers are in the off-state. The transistor is switched between on and off states by changing the gate voltage.

Most modern FETs are constructed using a combination of semiconductors, metals, and insulators; see Figure 2.1. Firstly, the semiconductor and its electrical characteristics determine the FET type. In an n-type FET, the majority charge carrier species are electrons, while holes dominate in p-type FETs.

Secondly, the metal layers form the electrical contacts that constitute the three terminals of the transistor. At the source and drain, metal-semiconductor junctions naturally create Schottky barriers with rectifying characteristics [11]. To facilitate current injection into the semiconductor, the conductivity beneath the contacts is increased, allowing low-resistance ohmic contacts to be formed and thereby reducing the contact resistance, R_c. In HEMT devices, the channel resistance is commonly described by the sheet resistance, R_{sh} (units $\Omega/\square$) [10]. Low values of both R_c and R_{sh} are essential for achieving high current drive capability and low on-resistance R_{on}, reducing conduction losses. Access resistances located

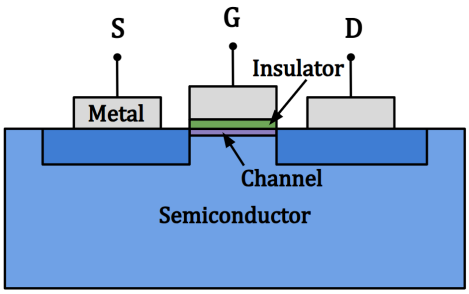


Figure 2.1: Schematic of a typical lateral FET with a metal-insulator-semiconductor (MIS) gate structure modulating the channel. The semiconductor is shown in blue. The channel is shown in purple. Regions with increased conductivity within the semiconductor are marked with a darker blue, and link the contacts with the channel region. Three metal contacts are drawn in light grey, with labelled terminals corresponding to the source (S), gate (G), and drain (D). An insulator is shown in green, forming the MIS junction with the metal gate contact and semiconductor below.

in the conductive regions between the source/drain pads and the channel can also contribute to the on-resistance.

The insulators are mainly used for forming metal-insulator-semiconductor (MIS) junctions at the gate or for device passivation and isolation [11]. Passivation and isolation of the devices are essential in reducing leakage currents, which can have detrimental effects on device performance.

FET gate structures are commonly realised either as a metal-semiconductor Schottky contact (MESFET) or as a MIS stack (MISFET, commonly referred to as a MOSFET when the insulator is an oxide) [11]. MESFETs may have low interface capacitance, given that a high-quality metal-semiconductor interface is formed. A lower capacitance can allow higher possible operation frequencies, but the absence of an insulator between the gate contact and the semiconductor can result in high gate leakage currents [10, 12]. In contrast, the insulator in a MIS gate structure limits the leakage but adds capacitance, lowering the maximum frequency of operation.

There are further subcategories of the FET that depend on the natural charge population state of the channel when no voltage is applied to the gate. FETs with high natural channel charge populations are referred to as depletion-mode devices. Depletion-mode devices are thus naturally in the on-state, and require a voltage applied to the gate to switch the device off. FETs with naturally non-populated channels are referred to as enhancement-mode devices. Enhancement-mode devices are instead naturally in the off-state, requiring a gate voltage to attract charge carriers to the channel to reach the on-state [11].

The metal-oxide-semiconductor FET (MOSFET) is widely studied and provides a valuable reference point for other classes of FETs. A set of equations

describing the current between the source and drain of the MOSFET, often called the drain current I_D, is given by

$$I_D = \begin{cases} 0 & V_{GS} \leq V_{th}, \\ \frac{1}{2}\mu C_{tot}\frac{W}{L}\left[2(V_{GS}-V_{th})V_{DS} - V_{DS}^2\right], & V_{DS} < V_{GS} - V_{th}, \\ \frac{1}{2}\mu C_{tot}\frac{W}{L}(V_{GS}-V_{th})^2, & V_{DS} \geq V_{GS} - V_{th}. \end{cases} \tag{2.1}$$

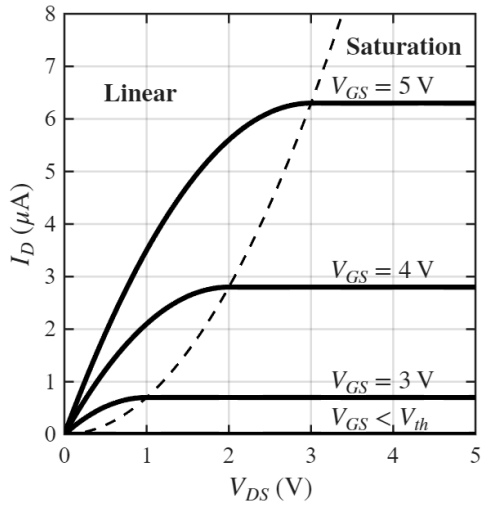
where μ is the mobility of the majority carriers in the channel when conducting, C_{tot} is the total capacitance present from the metal-oxide-semiconductor (MOS) gate structure, W and L are the device width and length, respectively, V_{GS} is the gate-source voltage, V_{DS} is the drain-source voltage and V_{th} is the threshold voltage [11]. The threshold voltage V_{th} refers to the gate voltage needed to turn the device on. More accurately, it signifies the transition from a depleted channel to a populated one. For $V_{GS} \leq V_{th}$, the device is turned off, modelled by $I_D = 0$. The two non-zero expressions in Equation 2.1 represent different operation regimes of the MOSFET. For the condition $V_{DS} < V_{GS} - V_{th}$, the device is said to be in the linear regime. For $V_{DS} \geq V_{GS} - V_{th}$, the device is in the saturation regime.

From Equation (2.1), when keeping V_{GS} constant, one retrieves the output characteristics, see Figure 2.2a. Similarly, when keeping V_{DS} constant, one retrieves the transfer characteristics, see Figure 2.2b. The current that the transistor conducts in its off-state, I_{off}, will in practice never be zero thanks to leakage mechanisms through the physical device, and is characterised using transfer curves for gate voltages where the device is deep in the off-state [13]. An on-current I_{on} can similarly be defined using a transfer curve for a gate voltage when the device is in saturation, giving another metric for device performance.

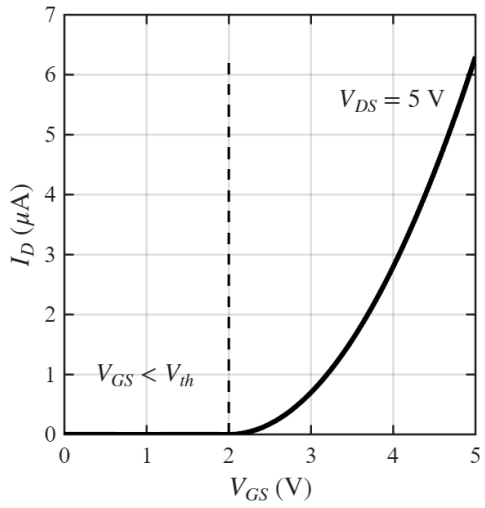
The on-resistance is defined in the linear region as $R_{on} = \frac{V_{DS}}{I_D}\big|_{V_{DS}\to 0}$ and is therefore extracted at low drain voltages. A low on-resistance reduces conduction losses, which is essential for lowering power consumption in FETs. The slope of the output curve in the saturation region can be used to define an output conductance as $g_d = \frac{\partial I_D}{\partial V_{DS}}$. Ideally, $g_d = 0$, but in practice, V_{DS} will continue to influence I_D slightly in the saturation regime, leaving g_d finite [11]. Furthermore, in short-channel devices where the gate length is scaled down, V_{DS} can lower the energy barrier needed for channel population, lowering the magnitude of V_{th}. This phenomenon is called drain-induced barrier lowering, and can be quantified from measured transfer I-V data as DIBL $= \frac{\Delta V_{th}}{\Delta V_{DS}}$.

A useful metric for a transistor's performance is the transconductance, defined by $g_m = \frac{\partial I_D}{\partial V_{GS}}$ [14]. The transconductance is mainly relevant to amplifier design, but can also be useful when finding an approximate V_{th} of a general FET, see Appendix B.

Other than g_m, DIBL, and V_{th}, the charge carrier mobility μ has a role in FET characteristics as exemplified for MOSFETs in Equation (2.1). μ describes the ease with which charge carriers move through the channel when an electric field is applied between the source and drain terminals [11]. A carrier will accelerate under an electric field $\mathcal{E}$, resulting in a current proportional to the drift velocity v_d. The drift velocity is limited by scattering events in the material, which counteract the acceleration from the applied field. At low-to-moderate field strengths, the



(a)



(b)

Figure 2.2: Two plots of MOSFET behaviour for parameters $\mu = 1400$ V/cm^2, $C_{tot} = 10$ nF, $W/L = 10$ and $V_{th} = 2$ V. a) The output characteristics for the device, showing output drain current versus drain voltage for different gate biases. b) The transfer characteristics for the device, showing output current versus gate voltage for a drain voltage of 5 V.

drift velocity is given by

$$v_d = \mu\mathcal{E} \tag{2.2}$$

The scattering events that limit v_d correspond to inelastic collisions between the charge carriers and the atoms, impurities, or defects in the crystal lattice of the material. High carrier mobility is achieved by carriers that minimise such collisions, and translates to lower on-resistance and higher on-currents. Successfully fabricating devices consisting of materials and material interfaces of high quality, containing a low amount of impurities and defects, is therefore a critical step when improving device performance.

At sufficiently high field strengths, carriers scatter frequently enough to limit their maximum achievable velocities. The saturation velocity v_{sat} represents an upper bound on the drift velocity, and limits the maximum carrier movement speed and the current a device can carry. This makes it a critical parameter for high-frequency and high-power applications, where high currents and switching speeds are desired. Equation 2.2 can be modified as

$$v_d = \frac{\mu\mathcal{E}}{1 + \mathcal{E}/\mathcal{E}_{\text{sat}}} \tag{2.3}$$

where $\mathcal{E}_{\text{sat}}$ is the field strength required to reach the onset of velocity saturation [11, 15].

A final consideration in FETs is the extent of gate control on the device. This is measured in the subthreshold regime of a FET, at gate voltages $V_{GS} < V_{th}$. In this regime, the drain current has an exponential relationship with the gate voltage [11]. The subthreshold slope is defined as

$$SS = \left(\frac{\partial \log_{10}(I_D)}{\partial V_{GS}}\right)^{-1} \tag{2.4}$$

A physical lower limit of the subthreshold slope for FETs is 60 mV/decade, meaning a 60 mV shift in V_{GS} leads to a 10 times change in I_D. This limit stems from the Fermi-Dirac distribution of the charge carriers in the semiconductor [11]. While some novel devices are able to beat the limit [16], the HEMTs relevant for this text stay above it [17, 18].

2.2 Power transistor requirements

One application area of FETs is found in converter technology within the field of power electronics. To illustrate, the circuit for a DC-DC step-down voltage converter, or buck converter, can be seen in Figure 2.3. The circuit is designed to create an output voltage V_{out} which is a set fraction of the input voltage V_{in} [19]. The fraction is decided by switching the switch S seen in Figure 2.3 on and off. The duty cycle d of the switching, being the fraction of time the switch is closed per switching period, relates the output voltage to the input voltage as $V_{out} = d \cdot V_{in}$.

The switch S is often implemented using FETs [3]. When the switch is open, achieved by an FET in the off-state, it will be subjected to a reverse voltage of

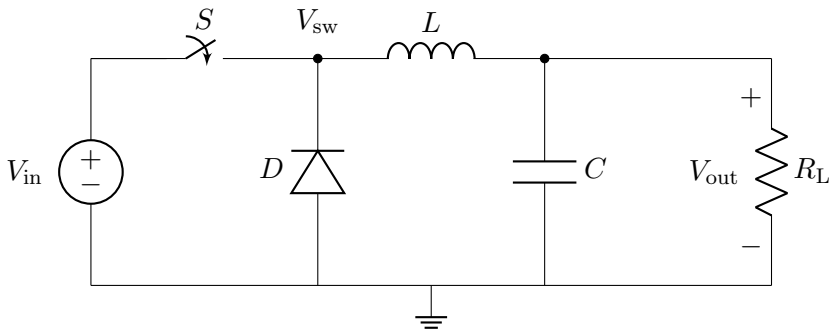


Figure 2.3: Schematic of a synchronous buck (step-down) converter. The transistor switch S toggles at high frequency to regulate the output voltage $V_{\text{out}} < V_{\text{in}}$. The switching node voltage V_{sw} alternates between V_{in} and ground. The inductor L and capacitor C form a low-pass filter, smoothing the output voltage, while the diode D provides a current path during the off-state.

$-V_{in}$. For the circuit to function correctly, the transistor is required to sustain this voltage in the off-state. For high-voltage circuits, the magnitude of V_{in} can be significant, motivating the need for the power FET to have a high off-state voltage tolerance, avoiding device breakdown. In semiconductors, the critical electrical field $\mathcal{E}_c$, marking the onset of breakdown, increases with the material's band gap E_g [3]. High band gap materials like GaN (E_g = 3.5 eV) and SiC (E_g = 3.26 eV) are thus preferred over the traditional FET material Si (E_g = 1.12 eV) when designing power FETs.

The passive circuit components C and L influence the physical realisation of the converter. If the modulation of the switch S is done at a high frequency, the passive components require a shorter time to store energy than for lower frequency modulation, allowing them to be scaled to smaller dimensions [19]. This reduction in component size is particularly valuable in converters that must simultaneously accommodate high currents. High-frequency operation of the FET switch is thus beneficial for converter designs.

The thermal conductivity λ and relative permittivity ϵ_r play roles in power device performance. A high thermal conductivity allows the material to dissipate heat easily, important for high-power applications where significant heat generation occurs from power losses [20]. The permittivity plays a role in parasitic capacitances, with a lower permittivity leading to a reduction of these. Lower parasitic capacitances allow for higher switching speeds and less dynamic power losses.

To benchmark a semiconductor material for power applications, the mobilities, critical fields, bandgaps, saturation velocities, inverse permittivities, and thermal conductivities of different candidates can be compared, see Figure 1.1a. Beyond pure material parameters, figures of merit are typically used. For performance at lower frequencies, the Baliga figure of merit (BFOM) is commonly used [3, 20], with higher BFOM values being better. The BFOM originates from the analysis of

unipolar power devices and quantifies the theoretical trade-off between conduction losses and breakdown capability. In its original derivation, the conduction losses are represented by the specific on-resistance $R_{on,\mathrm{sp}}$ associated with the device's current-carrying region. Although the physical origin of $R_{on,\mathrm{sp}}$ differs between device architectures, BFOM remains a useful metric for comparing the intrinsic suitability of semiconductor materials for power applications. The BFOM is expressed mathematically as

$$\mathrm{BFOM} = \epsilon\mu\mathcal{E}_c^3 \tag{2.5}$$

where ϵ is the permittivity of the semiconductor, μ the mobility and $\mathcal{E}_c$ the critical field for breakdown. From Eq. (2.5), the BFOM can be increased by using dielectrics with high relative permittivity, high mobilities, and significantly increased by increased resistance to breakdown. Again, this highlights the need for high bandgap materials over traditional Si. A plot can be constructed comparing the breakdown voltage and specific on-resistance corresponding to a constant BFOM. Materials with superior power-device performance exhibit higher achievable breakdown voltages for a given specific on-resistance, or equivalently lower specific on-resistance at a given blocking capability, as illustrated in Figure 1.1b.

For high voltage and high-frequency performance, Johnson's figure of merit (JFOM) is used for evaluation [21]. Mathematically, it is expressed as

$$\mathrm{JFOM} = \left(\frac{\mathcal{E}_c \cdot v_{sat}}{2\pi}\right)^2 \tag{2.6}$$

where $\mathcal{E}_c$ is the critical field and v_{sat} is the saturation velocity of the device. A high JFOM implies high voltage tolerance from the $\mathcal{E}_c^2$ contribution, as well as high-frequency operation from the v_{sat}^2 term. Materials which inherently have high $\mathcal{E}_c$ and v_{sat} values should be considered for high voltage and high-frequency power devices.

2.3 AlGaN/GaN high-electron mobility transistors

The high-electron mobility transistor (HEMT) is a type of FET that utilises the properties of certain semiconductor heterostructures to create an artificial, two-dimensional electron gas as the channel of the transistor. In HEMTs, a two-dimensional channel spatially confines the electrons in a quantum well, separating them from most of the impurities or defects in the material [9]. The probability of electrons losing momentum due to scattering is then greatly reduced. This, in turn, allows for higher mobilities and saturation velocities of the charge carriers. These resulting high carrier mobilities and saturation velocities make HEMTs attractive for high-frequency and high-power applications, both of which demand fast switching and high current handling capability.

Figure 2.4 shows a general diagram of a lateral HEMT. On top of the semiconductor substrate, one or more epitaxial semiconductor layers are grown to form the device structure. A heterostructure is created by growing a semiconductor layer with a wider bandgap, referred to as the barrier layer, on top of another semiconductor layer with a lower bandgap, commonly called the channel layer. Due to the band alignment at the heterointerface, a two-dimensional electron gas (2DEG)

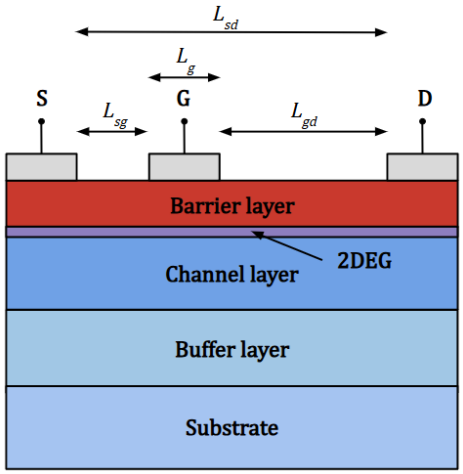


Figure 2.4: Diagram of a generalised, lateral HEMT. Locations of the source (S), gate (G), and drain (D) contacts, the barrier layer, 2DEG, channel layer, buffer layer, and substrate are indicated in the figure. The gate length L_g, the source-gate length L_{sg}, the gate-drain length L_{gd}, and the source-drain length L_{sd} are also indicated.

forms at or near the interface between the channel and barrier layers. This 2DEG acts as the high-mobility conduction channel of the HEMT. The channel layer rests on a buffer layer, which itself rests on the substrate. The insertion of a buffer layer, which blocks current leakage from the channel layer to the substrate, can assist in increasing the drain current of the transistor [22]. The metal source and drain contacts create Schottky contacts with the barrier layer. The gate contact can either be a metal-semiconductor (Schottky barrier) type, as indicated in the figure, or an MIS type. The length of the gate L_g, the source-gate length L_{sg}, the gate-drain length L_{gd}, and the source-drain length L_{sd} are important device design parameters that can influence the electrical properties of the HEMT.

Different types of HEMTs achieve a 2DEG in different ways. Of importance in this text is the polarisation-based 2DEG formation, used by AlGaN/GaN HEMTs [10, 23]. A two-dimensional quantum well is induced in the channel layer's conduction band close to the channel-barrier interface, where electron confinement subsequently creates the 2DEG. The quantum well is formed by the combination of inherent piezoelectric and spontaneous polarisations between the channel and barrier materials. Figure 2.5 shows charge and conduction band diagrams for a typical contacted AlGaN/GaN HEMT. For the AlGaN/GaN device, three interfaces contribute to the resulting band diagram: (1) the buffer/channel interface, (2) the channel/barrier interface, and (3) the barrier/contact interface.

Starting with interface (1), the GaN buffer and the channel layer on top are often grown in the $\langle 0001 \rangle$ (C) direction on a typical substrate of Si, sapphire, or SiC [10]. Furthermore, standard devices are grown as Ga-polar (+c) oriented, meaning the N-face is located at the bottom towards the substrate, while the Ga-

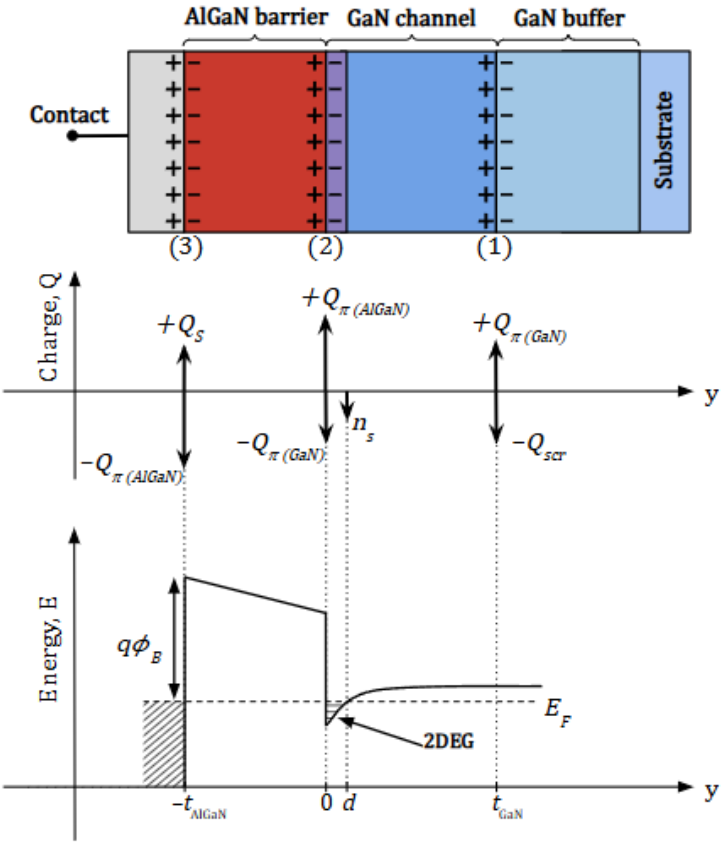


Figure 2.5: Contacted AlGaN/GaN HEMT structure with charges illustrated using arrows of varying magnitude, and the resulting conduction band diagram. Three interfaces are marked: the buffer/channel interface (1), the channel/barrier interface (2), and the barrier/contact interface (3).

face is on top. The N-face has the net surface polarisation charge of $+Q_{\pi\ (\mathrm{GaN})}$, while the Ga-face has charge $-Q_{\pi\ (\mathrm{GaN})}$. The positive charge on the N-face is screened by charges within the substrate $-Q_{scr}$.

At interface (2), the remaining $-Q_{\pi\ (\mathrm{GaN})}$ of the GaN channel faces the positive AlGaN polarisation charge $+Q_{\pi\ (\mathrm{AlGaN})}$. The magnitude of AlGaN polarisation charge is greater than that for GaN, $|Q_{\pi\ (\mathrm{AlGaN})}| > |Q_{\pi\ (\mathrm{GaN})}|$. Thus, the net charge at interface (2) is positive, causing an electrical field to be induced [10, 23]. The resulting band bending on the GaN side corresponds to an attraction of electrons n_s, which are located within a triangular quantum well, forming the 2DEG. These electrons neutralise the remaining positive charge from interface (2). The 2DEG occupies energy states below the Fermi level E_F and extends to a distance d from the GaN/AlGaN interface.

Interface (3) houses a Schottky barrier of height ϕ_B between the metal contact and the semiconducting AlGaN. The charge at the interface is equal to the difference between surface charges $+Q_s$ from the contact and $-Q_{\pi\ (\mathrm{AlGaN})}$, causing a negative net charge in the Schottky interface region for thin AlGaN layer thicknesses. For gate contacts, the Schottky barrier is intentionally kept to suppress gate leakage, while it is reduced to form ohmic contacts at the source and drain.

An analysis of the energy landscape can be done to get an equation for the 2DEG electron sheet density n_s dependence on the applied gate voltage V_G, found to be

$$n_s(V_G) = \frac{\left(Q_{\pi\ (\mathrm{AlGaN})} - Q_{\pi\ (\mathrm{GaN})}\right) t_{\mathrm{AlGaN}} + \epsilon \left(V_G - (\phi_{B,\mathrm{gate}} - \frac{\Delta E_c}{q})\right)}{q(t_{\mathrm{AlGaN}} + d)} \tag{2.7}$$

where t_{AlGaN} is the AlGaN layer thickness, ΔE_c is the difference in conduction band energy between AlGaN and GaN, and d is the extent of the 2DEG region [9, 10]. A higher gate voltage V_G will thus lead to an increase in the channel electron concentration n_s, increasing conductivity.

2.4 Common modifications of AlGaN/GaN HEMTs

A typical side effect caused by the AlGaN/GaN HEMT 2DEG is that the HEMT naturally operates in depletion-mode operation. The transistor threshold voltage is thus negative, $V_{th} < 0$, as a negative bias is required to deplete the channel of electrons and turn the device off.

For applications requiring enhancement-mode operation, a shift of V_{th} towards positive voltages is required. One commonly used approach for this involves modifying the gate structure by recessing the AlGaN barrier beneath the gate, often combined with a MIS gate structure for added gate leakage suppression [9, 24, 25], see Figure 2.6. Recessing reduces 2DEG charge density under the gate, which in turn helps shift V_{th} toward more positive values. The reduced 2DEG density can, however, lead to a reduction in the maximum on-currents of the device. An added benefit of the recessed MIS structure is the possibility of improving forward voltage breakdown, being a function of the chosen insulator thickness [9]. A downside is the restricted choice of possible insulators and the great quality required of the deposited insulators to avoid charge traps and reduced 2DEG mobility.

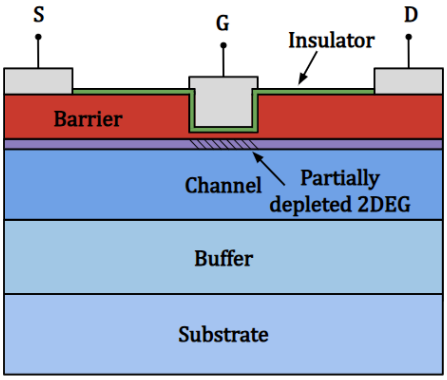


Figure 2.6: Diagram of a lateral MIS-HEMT with recessed gate. The barrier beneath the gate contact has been recessed, with an insulator inserted across the surface. A MIS junction is created between the metal gate contact, the insulator, and the semiconducting barrier layer.

The 2DEG sheet concentration can be increased by inserting a larger conduction band difference ΔE_c at the 2DEG interface, see equation (2.7) above. Inserting a thin AlN interlayer between the AlGaN barrier and GaN channel will increase ΔE_c, as the conduction band difference between AlN and GaN is larger than that of AlGaN and GaN. AlN is more amenable to growth on GaN than AlGaN, reducing the probability of defect formations around the 2DEG, further increasing the possible channel conductivity [26].

The source and drain contacts can be modified to provide improved device performance. Lowering the contact resistance allows a higher maximum current extracted from the device. A recess of the AlGaN barrier below the source and drain contacts can be done to improve electron flow from the contacts to the 2DEG, thus reducing the resistance. A deep recess has been shown to decrease the contact resistance, and there is a strong contact annealing temperature dependence for achieving an optimal low contact resistance [27]. Another tactic can be employed at the contacts to improve the breakdown of the device. Adding field plates can assist in smoothing the electric field distribution within the device at high voltages [9]. A smoothed distribution will result in a lower overall peak field, thus extending the voltage range of the device until the field reaches $\mathcal{E}_c$.

An undesired property of GaN in practice is unintentional doping (UID) during growth [28]. The doping can increase leakage current in the transistor off-state, which increases power losses and decreases the safety margin for current breakdown. Neutralisation of the unintentional doping can be achieved using ion implantation of, for example, C, N, or Fe ions [29]. Leakage between devices is also an issue, and can be mitigated by performing a deep MESA isolation etch in the intermediate regions, exposing the mesa-like layering of the material stack, or by ion implantation [30].

The exposed AlGaN surface seen in Figure 2.4 can cause premature device

breakdown and influence the on-resistance R_{on}. This breakdown is induced by the presence of charge traps on the AlGaN surface, which can assist in depleting the channel of charge, reducing 2DEG conductivity. To solve this issue, a surface passivation layer can be deposited on top of the AlGaN barrier, neutralising surface states [9].

Test structures can be implemented on a sample to extract certain parameters during or after processing. One such test structure is the transmission line model (TLM) structure. TLM structures consist of arrays of metal pads placed on top of the sample with varied spacing. Measurements between TLM pads are used for quantifying the contact resistance, breakdown voltage, and conduction through the contacts and the underlying semiconductor [31]. Four-point probe measurements are preferred for TLM structures to minimise the influence of the probe resistances while measuring.

2.5 AlGaN/GaN HEMT fabrication steps

The material stack used to produce AlGaN/GaN HEMTs in this work can be seen in Figure 2.7. Schottky and MIS HEMTs are to be fabricated on the same sample in different regions, with geometric and field plate variations. The stack is grown on a semi-insulating (SI) GaN substrate. The GaN buffer is initiated by growing a 150 nm thick GaN layer at low temperature (LT) for enhanced material quality. A 1 µm thick GaN channel layer is subsequently grown on top, and is characterised by unintentional doping. A 1 nm thin AlN interlayer is grown on the UID-GaN, with a 15 nm $Al_xGaN_{1-x}N$ barrier layer on top, with Al content of $x \approx 0.15$. The 2DEG is located at the interface between the UID-GaN layer and the AlN interlayer. Finally, a 1-2 nm GaN cap is grown on top of the AlGaN layer to prevent oxidation.

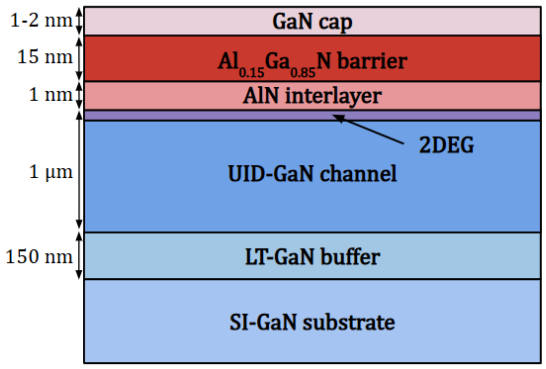


Figure 2.7: Material stack used in the HEMT fabrication described in this work. In subsequent figures, the SI-GaN substrate as well as the marked 2DEG region will be omitted for simplicity.

The first step in HEMT processing involves device isolation. A MESA etch is performed to isolate the devices from each other. The MESA etch is followed by

a source/drain contact recess. A schematic illustrating the MESA and recess-etch outcomes on the material stack is shown in Figure 2.8.

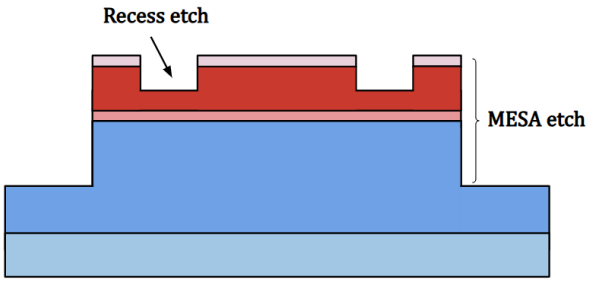


Figure 2.8: Illustration of the first steps in the HEMT processing, using the material stack from Figure 2.7. A MESA etch has been performed, illustrated by pits formed at either end of the stack. The recess etch is shown to have etched away the GaN cap and partially into the AlGaN barrier in two locations, defining the source and drain.

After recess etching, Ti/Al/Ni/Au contacts are deposited. After deposition, the metal contact is annealed by heating rapidly to a high temperature for a short time, see Figure 2.9.

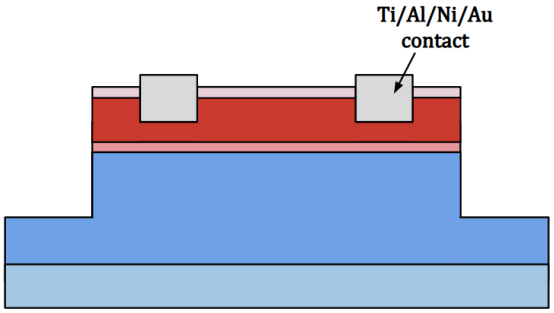


Figure 2.9: Metal contacts added to the recessed regions in the material stack.

Additional device isolations were performed by ion implantation in the etched MESA regions to neutralise the UID-GaN channel layer and underlying buffer, followed by an additional MESA etch around the devices for added isolation, see Figure 2.10.

Following the added isolation, a SiN_x passivation layer is deposited on top of the devices, see Figure 2.11.

A gate recess is performed next at the location of the planned gate contacts,

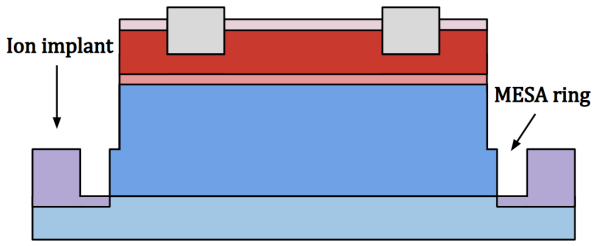


Figure 2.10: The material stack after additional isolation by ion implantation and addition of a ring trench around the device, ending at the LT-GaN buffer layer.

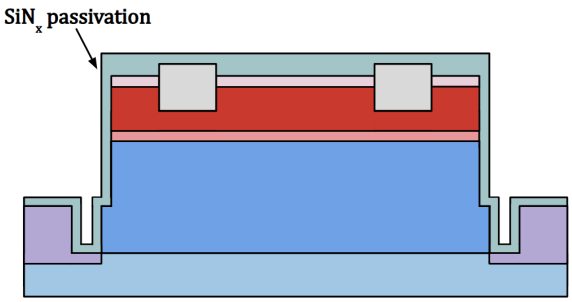


Figure 2.11: A SiN_x passivation layer has been deposited over the surface of the entire sample.

etching through the SiN_x passivation, GaN cap, and partly through the AlGaN barrier. After etching, a thin dielectric of Al_2O_3 is grown above the entire sample. The Al_2O_3 is subsequently removed over areas where Schottky devices are planned, and kept in regions for MIS devices. A gate metal of Ni/Au is then deposited in the gate regions, see Figure 2.12.

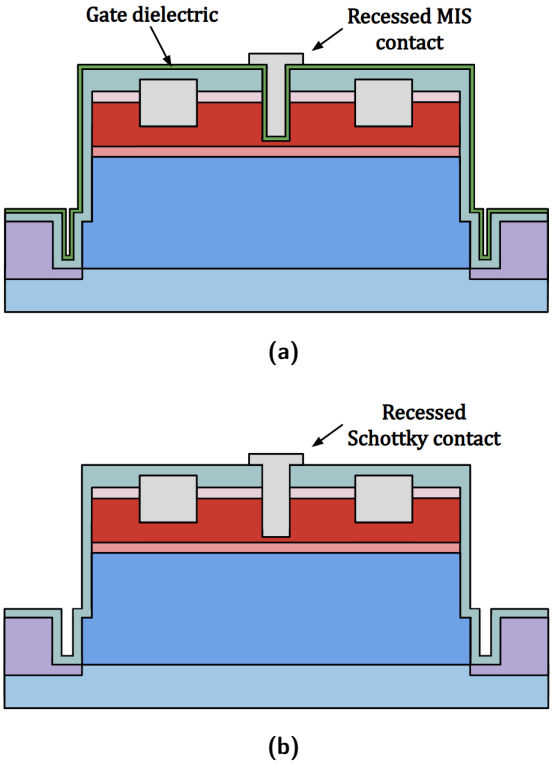
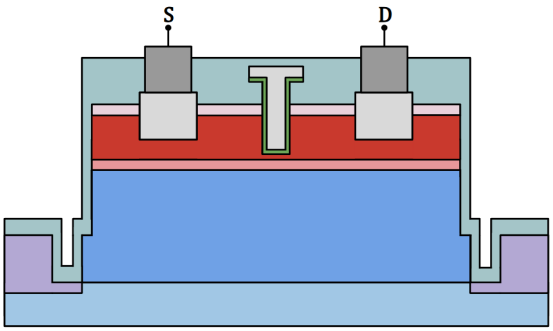
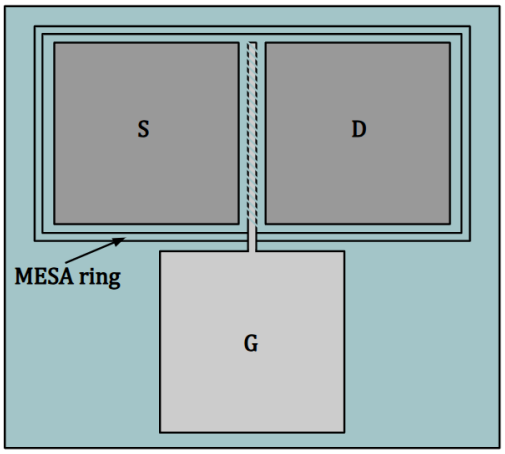


Figure 2.12: A gate recess has been performed, followed by Al_2O_3 dielectric deposition. a) A gate metal is deposited on top of the dielectric, forming a MIS junction. b) The deposited dielectric is removed across the device region before metal deposition, leading to a Schottky contact.

Excess Al_2O_3 gate dielectric in regions beyond the MIS gate structures is etched away, and a second SiN_x passivation layer is grown on top of the devices. In a subsequent step, vias for the source and drain are formed by etching through the SiN_x stack above the contact regions. Four contact variations of the device are used: a source field plate design, a gate field plate design, a gate and drain field plate design, and finally a design without field plates with simple source and drain contacts. The latter design, free of field plates, can be seen in Figure 2.13, showcasing a finalised MIS-HEMT device structure. The field plate variations used for a portion of the processed devices can be seen in Figure 2.14.



(a)



(b)

Figure 2.13: Two perspectives of a processed MIS-HEMT device. a) A cross-sectional view of the MIS-HEMT with field plates. The gate contact is not shown in the figure. b) The top view of the HEMT, showcasing the location of the source, drain, and gate pads relative to the gate finger. The gate finger is shown as striped when located beneath the second passivation layer, and fully coloured when exposed to air. The location of the MESA isolation ring is also showcased.

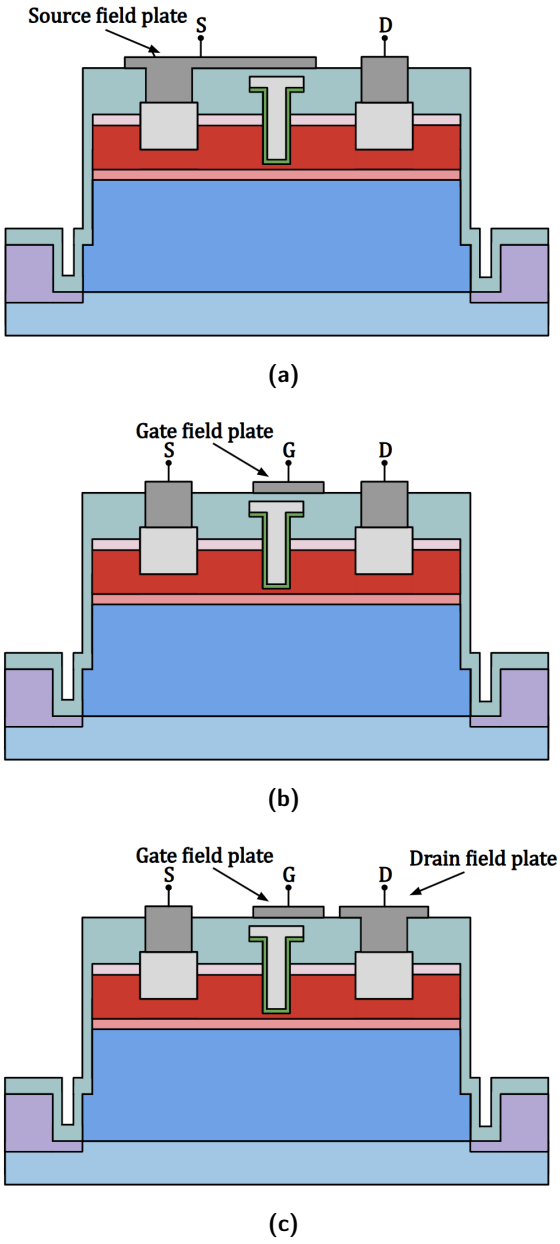


Figure 2.14: Field-plated HEMT designs. a) A source field plate design, with the source pad extending over the buried gate contact. b) A gate field plate, situated on top of the buried gate finger and extending towards the drain. c) Gate and drain field plates.

Chapter 3

Process optimisation and device fabrication

An AlGaN/GaN HEMT material stack was grown at Linköping University of Technology, with UID-GaN layer thickness of 1 µm, AlN interlayer thickness of 1 nm, and $Al_{0.15}Ga_{0.85}N$ buffer thickness of 15 nm, see Figure 2.7 above. The material stack demonstrated favourable sheet resistance, mobility, and electron sheet density values compared to the literature [32, 33], see Table 3.1. The following

Table 3.1: Reported material stack properties.

R_{sh} ($\Omega/\square$)	μ (cm^2/Vs)	n_s (cm^{-2})
154	944	$4.48 \cdot 10^{13}$

sections cover a description of the main processing steps used for the HEMT device fabrication, as well as the optimisation of two main HEMT processing steps. This is followed by a description of the HEMT fabrication.

3.1 Clean-room processes

Throughout the process of optimisation and device fabrication, a plethora of clean-room equipment was used. The Lund Nano Lab clean room was utilised for this degree project. The equipment used for the project, as well as some commonly repeated process parameters, is described below.

One of the important processes in semiconductor fabrication is sample cleaning. In steps mentioning solvent cleaning, the sample is cleaned in a solution of acetone using low-power sonication, followed by cleaning in IPA using low-power sonication, and then dried using a N_2 gun. Later, it is baked to remove excess solvents, making the surface ready for a clean and consistent lithography process.

Lithography involved spinning the resists based on post-process requirements. For steps requiring feature definition via etching, either ma-N 440, S1813, or S1828 resist was used. For steps involving metal deposition, a bilayer involving LOR10B and S1813 was used. All the resists were spun at 4000rpm for 60s, followed by a bake. The bake was done at 95°C for 3 minutes for ma-N 440 resist, at 120°C for 5 minutes for S18XX resists, and at 180°C for 5 minutes for LOR10B resist. The exposure of the resist was performed using the Heidelberg Instruments MLA150 maskless aligner. A laser of wavelength 375nm was used for the exposure, and

the layer was exposed using a suitable dose and defocus parameter. The resist development was carried out in suitable developer solutions. The ma-N 440 resist was developed in the ma-D 532/S developer, while the S18XX resists and the LOR10B resist were developed in the MF-319 developer for the required time and then rinsed in de-ionized (DI) water to stop the development. The sample is then dried using a N_2 gun. To ensure the complete removal of resists from developed regions, ashing is performed as needed using a PlasmaPreen microwave plasma ashing system.

Etching was carried out using three different systems available at the Lund Nano Lab: two inductively coupled plasma reactive ion etching (ICP-RIE) tools and a conventional reactive ion etching (RIE) system. A Cl-based Plasma-Therm APEX SLR ICP-RIE was used mainly for GaN etching, and both an F-based Plasma-Therm APEX SLR ICP-RIE and a Trion T2 RIE etcher were used for SiN_x etching.

Post-etching processes for samples involved stripping the resist layer, which was used as a hard mask. This was done by heating the sample in acetone at 60°C for 5 minutes or until the resist is completely removed. The sample is then rinsed in IPA and dried using a N_2 gun.

Metal evaporation is done using a Temescal FC-2000 E-beam evaporator, and metal sputtering is done using an AJA Orion 6 sputterer. Metal lift-off was done in a solution of remover 1165 heated to 80°C for 10 minutes or longer if needed. The sample is later rinsed in IPA after ensuring that the metal has been removed from the desired regions.

Dielectric layers for passivation and ion-implantation masks were done using plasma-enhanced chemical vapour deposition (PECVD) in a SPTS DELTA LPX APM system. The dielectric for the MIS-gated structures was deposited using an atomic layer deposition (ALD) technique using a Beneq TFS 200 system.

For metrology purposes, a J. A. Woollam RC2 ellipsometer was used to measure the thicknesses of the deposited insulator layers and dielectrics. A Bruker Dektak XT Stylus profilometer was utilised to measure the profile of the samples after different steps when necessary. A Zeiss GeminiSEM 500, equipped with an EDS detector, was used for imaging and material characterisation.

3.2 Process optimisation

The following section will cover two important process optimisations for the fabrication of the AlGaN/GaN HEMTs.

3.2.1 Recess etch optimisation

A low-power etching recipe is recommended for recess etch during ohmic contact formation in AlGaN/GaN HEMTs to minimize plasma-induced damage to the AlGaN barrier and the underlying 2DEG channel. Gentle etching helps preserve carrier mobility and sheet conductivity while reducing defect generation, surface roughness, and trap formation. This is important for achieving low contact resistance and promotes good device performance and reliability [34].

Three variations of source/drain recess etches were planned for the HEMT devices to facilitate the formation of ohmic source/drain contacts. One region was planned to remain unetched, while the other two were to be recessed by 3 nm and 5 nm, respectively.

Optimisation tests were undertaken using dummy samples of GaN on which S1813 resist was spun, and the MLA was used to pattern the sample with test structures. Development was done in pure MF-319 for 17 seconds, and the test patterns were confirmed using optical microscopy.

The recess etching was done using the Cl-based ICP-RIE etcher. The recipe used Cl_2/BCl_3 gases with a flow rate of 18 sccm/2 sccm, respectively. The chamber pressure was maintained at 5 mTorr, and the RF power was planned to be varied between 2 and 6 W, with the intention of causing a mild etching. After preconditioning the etcher, etching was done at 6 W on a prepared sample for 5 minutes.

The profilometer was used to compare the height difference between etched and non-etched features on the dummy samples. For the 6 W etched sample, negligible etching could be detected. The following etching trials then varied the RF power between 9, 12, and 15 W instead of the previously planned values. Using the profilometer, it was determined that the 12 W recipe was most suitable for the recess step, having an etch rate of ≈ 2 nm/minute.

3.2.2 Dielectric optimisation

SiN_x is commonly used as a passivation layer in GaN HEMTs because it helps suppress surface states and current collapse by reducing charge trapping at the AlGaN surface [35]. Al_2O_3 is widely used as a gate dielectric in MIS-HEMTs due to its high dielectric constant, good insulating properties, and high breakdown strength [36].

To determine the processing conditions needed for good-quality SiN_x and Al_2O_3, trials were conducted using the PECVD tool for SiN_x and the ALD tool for Al_2O_3. SiN_x was deposited onto two Si samples: one was deposited using NH_3 gas; the other without. Both the depositions were done at a temperature of 300°C. A third sample, deposited using low-pressure chemical vapour deposition (LPCVD) at Chalmers Nanofabrication Laboratory, was available from previous experiments within the group and was used as a benchmark. Al_2O_3 was deposited on a Si sample using ALD at a temperature of 300°C. Tetra-methyl aluminium (TMA) and H_2O were used as precursors for the ALD deposition.

The PECVD SiN_x samples and Al_2O_3 samples were etched in a 1:5 solution of HF. The initial and final thicknesses of the deposited dielectric layers were determined using the ellipsometer. After etching, the change in thickness was noted, and an etch rate was determined. A similar etch test was also done for the benchmark LPCVD SiN_x. The recorded etch rates can be seen in Table 3.2 below.

To perform electrical characterisation, contact pads were deposited to form MIS capacitor structures. The deposition was done using LOR10B/S1813 lift-off. A stack of Ti/Au with 10/100 nm thickness was deposited using the metal evaporator. After lift-off, optical microscopy was used to visually verify the result. A similar metal stack was deposited on the backside of the substrate, contacting

Table 3.2: Measured wet etch rates and electrical characteristics of SiN_x and Al_2O_3 layers. Unsuccessful electrical characterisation is marked with "-".

Material	Etch rate (nm/s)	Breakdown field (MV/cm)	Flat-band voltage (V)
LPCVD SiN_x	0.03	14.3	-7.6
$PECVD_{no\ NH_3}$ SiN_x	0.88	14.5	-7.8
$PECVD_{NH_3}$ SiN_x	2.05	-	-
Al_2O_3	4.7	8	-1.6

the Si.

A two-point measurement between the top oxide part and the bottom of the sample was carried out in a TS2000-SE Semi-auto probe station at the Department of Electrical and Information Technology at LTH, with characterisation performed by a Keysight B1500A Semiconductor Device Parameter Analyser. Device breakdown and capacitance-voltage (C-V) measurements were performed. The outcome from the C-V and I-V results can be seen in Table 3.2. The C-V measurements on the NH_3-based SiN_x layer were unsuccessful due to the layer being too thick (≈200 nm) and the resulting capacitance being too small. The LPCVD benchmark and the PECVD SiN_x deposited without NH_3 showed similar flatband voltages. In contrast, the Al_2O_3 showed a smaller flatband voltage, indicating that it is more suitable for the MIS gate than SiN_x. The I-V results revealed a larger breakdown field for the SiN_x samples than reported in the literature [37, 38]; this was later found to be due to traces of SiN_x being on the backside of the Si test samples after deposition, resulting in a higher breakdown voltage for the SiN_x samples. This error was mitigated in the Al_2O_3 samples by including an acid wet-etch step before backside metal deposition.

3.3 Device fabrication

The fabricated MIS-HEMT schematic can be seen in Figure 2.13 above. The steps taken to go from the given material stack to the finished devices are described below.

3.3.1 MESA isolation trench formation

The first step to go from the material stack to the finished devices was the definition of MESA isolation trenches, refer to the MESA etch in Figure 2.8.

The sample was cleaned with a 1:1 H_2SO_4 to H_2O_2 pirhana solution for 5 minutes, then cleaned in acetone and IPA for two minutes. To pattern the regions for isolation trench etching, S1813 was coated on the sample and exposed in the MLA. The exposure dose was 700 mJ/cm^2. The sample was developed in a 6:1 MF-319:H_2O solution for 60 seconds. O_2 ashing was done at 5 mbar in the dedicated plasma asher for 30 seconds.

Isolation trench etching was carried out in an Oxford PlasmaPro 100 Cl_2 ICP-RIE etcher at Chalmers using a GaN etching recipe with 18:2 Cl_2:BCl_3 at 150 W RF power, 300 W ICP power, and at 5 mTorr pressure. The recipe had a GaN etch rate of approximately 202 nm/minute. The targeted MESA trench depth was 200 nm; thus, a 1-minute etch was carried out with the recipe. After etching, the S1813 resist was stripped, and the etch depth was verified using the profilometer. Leakage current measurements after etching revealed high leakage, necessitating a further isolation etch. Cleaning and patterning followed the same steps as above, and etching was performed in the Lund Nano Lab using a Cl-based ICP-RIE with a recipe of 18:2 Cl_2:BCl_3 at 120 W RF power, 150 W ICP power, and 5 mTorr pressure. This recipe has a GaN etch rate of approximately 315 nm/minute, and to etch an additional 300 nm into the trenches, a 57-second etch was carried out with the recipe.

3.3.2 Source/drain ohmic contact formation

After MESA trenches had been achieved, contact recess and contacts deposition were performed, refer to Figures 2.8 and 2.9, respectively. Three variations of the recessed feature depths were targeted in different regions of the sample: regions with no recess, regions with a 3 nm recess, and regions with a 5 nm recess.

Three separate processing steps were needed for the metal deposition in the three planned regions with varying recess depth, with two of the three requiring a recess etch before metal evaporation. The 5 nm recess was performed first, for which the sample was prepared for metal lift-off by coating it with LOR10B and S1813 resist. MLA exposure with a 700 mJ/cm^2 dose and 6:1 MF-319:H_2O development followed. Development was done for 60 to 75 seconds, until the features were visibly developed and a sufficient undercut, above 1 µm in width, was visible in an optical microscope.

A recess etch was done after the resist development. The etch used the optimised etch recipe found during the process optimisation, with an etch rate of approximately 2 nm/minute. To achieve a 5 nm recess, a 150-second etch was carried out. After etching, the resulting etch depth was verified using the profilometer on a dummy sample.

Before metal deposition, a digital etch using a 10:1 H_2O:HCl solution for 60 seconds was done to strip potential surface oxidation from the recessed structures, with the intention of reducing the contact resistance. After the digital etch, the sample was transferred to the metal evaporator, and a metal stack of Ti/Al/Ni/Au, with thicknesses of 30/100/40/240 nm, respectively, was deposited. After deposition, lift-off was carried out, and the resulting structures were verified using an optical microscope. These steps were then repeated for the 3nm recess variation. The 0 nm recess was similarly achieved the same way, without the recess etch step.

To reduce the contact resistance, an anneal of the metal contacts was performed in a UniTemp RTP-1200-100 annealer. The annealing was done using N_2 gas at 800°C and held for 30 seconds. After annealing, optical microscopy was performed, which indicated strong signs of surface roughness in the contacts, possibly impacting electrical performance. Contact resistance and sheet resistance were measured on TLM structures using the deposited metal contacts to verify

this observation, see Figure 3.1. Increased recess depth was observed to lead to a

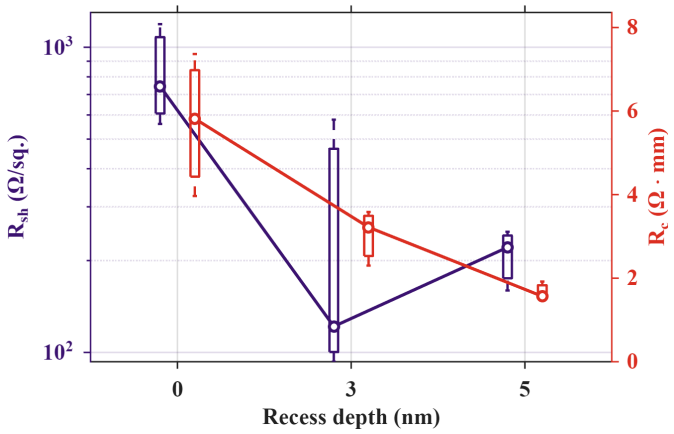


Figure 3.1: Measured contact and sheet resistance for the TLM structures after annealing.

decrease in contact resistance. However, for all measured TLMs, the magnitude of the contact resistance surpassed a desired value of 1 Ω·mm, implying that the annealing was partially unsuccessful [27]. Furthermore, the sheet resistance showed strong variations with the recess depth, indicating that the etch recipe may cause surface roughness. Insufficient etching of the GaN cap could influence the 2DEG, thus leading to unexpected R_{sh} fluctuations with recess depth, most notable for the 0 and 3 nm recess depths. The non-ideal annealing could also be a factor leading to variations in the sheet resistance. An attempt to rectify the issue with the annealed contacts was made, and can be found in Appendix A.

3.3.3 Ion implantation and MESA ring formation

The leakage between TLM pads isolated by MESA trenches was found to be high, despite a total recess depth of ≈ 500 nm. To further reduce leakage, ion implantation was employed. The implantation dose was adjusted to neutralise the unintentional background doping in the exposed UID-GaN layer in the etched MESA valleys between devices. Before implantation, a 1 µm SiN_x implantation mask was deposited on the sample surface using PECVD, with a recipe including NH_3. After PECVD, ellipsometry was used to find the actual thickness, which was 1.07 µm. The sample was then coated with ma-N 440 resist, exposed using MLA at a dose of 2000 mJ/cm^2, and subsequently developed in ma-D 532/S solution to define a pattern for the implantation mask. The F-based ICP-RIE was used to transfer the pattern into the SiN_x mask by etching down to the GaN surface.

Implantation of N^{2-} ions was done at Uppsala Tandem Laboratory. Three implantation energies were used, 30, 100, and 160 keV at doses of $3.3{\cdot}10^{12}$, $6.7{\cdot}10^{12}$, and $1.2 \cdot 10^{13}$, respectively. After implantation, the leakage was re-measured, see Figure 3.2. It was concluded that the implantation had not reduced the leakage by more than an order of magnitude. Further MESA isolation was concluded as a

possible solution. The SiN_x implantation mask was removed using a wet etch in a 1:5 HF: H_2O solution. The SiN_x etch rate was approximately 2.05 nm/s, and the 1.07 µm thick mask was etched for 8 minutes and 42 seconds. Scanning electron microscopy (SEM) with energy-dispersive X-ray spectroscopy (EDS) was used to characterise the sample surface and ensure no SiN_x mask remained.

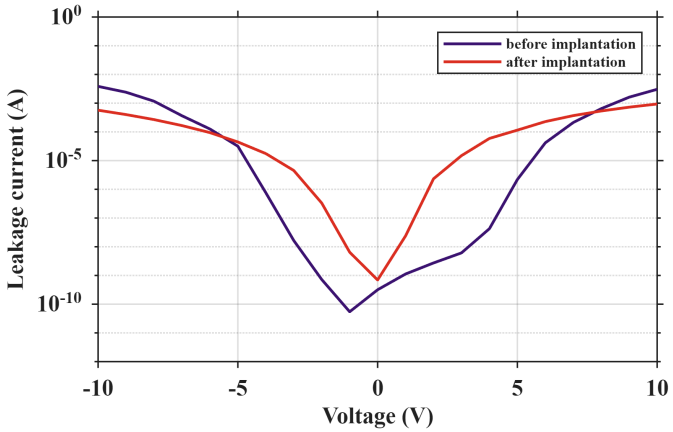


Figure 3.2: TLM MESA leakage measurements from before (purple) and after (red) the ion implantation.

An additional 500 nm deep rings were formed in the sample, refer to Figure 2.10. To prepare the sample for the additional MESA isolation, it was coated with S1813 and exposed with the MLA with a 600 mJ/cm^2 dose, then developed for 55 seconds in 6:1 MF-319:H_2O. Before etching, O_2 plasma ashing was carried out at 5 mbar for 30 seconds. Etching was done in the Cl-based ICP-RIE. To etch 500 nm, the sample was etched for 1 minute and 35 seconds. Profilometry confirmed a total MESA trench depth of 1 µm in the etched ring regions.

3.3.4 First passivation deposition

After the source/drain contact deposition and additional leakage reduction structures, a passivation layer of SiN_x was deposited, refer to Figure 2.11. The sample was prepared for deposition by an acid clean in a 1:5 HF: H_2O solution for 10 seconds, followed by a 5-minute bake at 120°C. Passivation deposition was done using PECVD. Two layers of passivation were deposited: a first layer targeting 10 nm of NH_3-based SiN_x was deposited, followed by a 55 nm SiN_x layer without NH_3. Si wafers were placed in the PECVD chamber along with the main sample to deduce the real deposited layer thickness using ellipsometry for both depositions. The first layer was found to have a thickness of 8.8 nm, and the second layer 57.6 nm, with a total thickness of 64.2 nm.

3.3.5 Gate Definition

After the first passivation deposition, the gate was defined, see Figure 2.12. As in the source/drain contacts, three recess variations were planned for the gate:

one region with no gate recess, one with a 6 nm recess, and one with an 11 nm recess. The total height variation of the sample was around 1.5 µm due to the additional MESA trench etching and the rough source/drain pads post-anneal. To successfully cover the sample with resist, a bilayer of LOR10B and S1813 was employed. After bilayer coating, the sample was exposed in the MLA using a 700 mJ/cm^2 dose and developed in 6:1 MF-319:H_2O for 55 seconds. Three gate lengths were transferred to the resist, being 0.5 µm, 1 µm, and 2 µm. The 0.5 µm features were not properly developed, and to avoid over-development of the remaining larger features, O_2 ashing for 2 minutes at 5 mbar was performed, followed by a 20-second dip in the development solution. Optical microscopy revealed that development was successful, with the side effect of the targeted 500 nm gate features ranging between 600 and 700 nm.

The patterning was succeeded by etching of the 64.2 nm SiN_x passivation layer in the developed regions. Two etch steps were used for added control. The first etch step was done in the Trion etcher, using a recipe with a 5/50 sccm O_2/CHF_3 gas at 75 W RF power and 20 mTorr pressure. The SiN_x etch rate of the recipe depended on deposition conditions. For SiN_x deposited without NH_3, the etch rate was approximately 0.88 nm/second. For SiN_x with NH_3, it was around 1.14 nm/second. The target depth was 58 to 60 nm, and the sample was etched for 70 seconds with the recipe, removing the SiN_x without an NH_3 layer. After the dry etch, a wet etch using a 1:5 HF:H_2O solution was performed. The solution etched SiN_x deposited with NH_3 at approximately 2.05 nm/second. The target depth was between 4 and 6 nm. Si test samples with the same SiN_x passivation stack were etched alongside the main sample, and ellipsometry was performed on these to verify SiN_x removal.

After SiN_x etching, the AlGaN barrier layer was etched using the Cl-based ICP-RIE. The optimised recess recipe with a 2 nm/minute etch-rate was used, and the AlGaN was etched for 5 minutes and 30 seconds, targeting an 11 nm gate recess. The etch depth was verified using the profilometer. After sample cleaning, the same processing as above was done for a different region on the sample, this time targeting a 6 nm gate recess by etching for 3 minutes in the Cl-based ICP-RIE.

Following the gate recess was the gate insulator deposition. The sample was prepared using an acid clean in a 1:10 HF: H_2O solution for 5 seconds, followed by a 5-minute bake at 120°C. A 10 nm Al_2O_3 gate oxide was then deposited using ALD. Si test samples were placed in the ALD chamber to assist in measuring the oxide thickness by ellipsometry. The measured Al_2O_3 thickness on the actual sample was found to be 11.2 nm.

Some devices were planned to remain as Schottky HEMTs, not having a MIS-structure at the gate. For the devices where the MIS structure would remain, the gate contacts were deposited directly using a LOR10B/S1813 based lift-off resist coating, exposed in the MLA with a 600 mJ/cm^2 dose, and developed for 55 seconds in 6:1 MF-319:H_2O. The gate metal was deposited using the sputterer, with a Ni/Au metal stack with targeted thicknesses of 30 and 100 nm, respectively. After resist lift-off, the actual deposited height was found to be 115 nm, using profilometry. For the Schottky HEMTs, the LOR10B/S1813 resist coating and patterning were performed. Before metal sputtering, the Al_2O_3 across the Schottky sample areas was removed by dipping the sample in a 1:10 HF:H_2O solution for 5s, and

a subsequent bake for 5 minutes at 120°C. The same Ni/Au gate metal was then deposited on the sample, followed by a lift-off process.

3.3.6 Second passivation deposition and via etching

A second passivation was deposited above the exposed gate metal, and vias were defined, see Figure 2.13a. A 10-second acid dip in 1:10 HF: H_2O was performed to remove excess exposed Al_2O_3, followed by a 5-minute bake at 120°C. PECVD of a first 10 nm targeted SiN_x layer using NH_3 was performed, followed by a 65 nm targeted SiN_x deposition without NH_3 assistance.

For the via definitions, the sample was cleaned in acetone and IPA, then pre-baked at 180°C for 5 minutes. For patterning of the gate via openings, S1828 resist was spun, then the sample was post-baked at 120°C for 5 minutes. The resist was exposed in the MLA with a 600 mJ/cm^2 dose, then developed for 55 seconds in 6:1 MF-319:H_2O. After patterning, the second SiN_x passivation layer was etched in the Trion etcher. The same recipe used above for the passivation removal before gate recess was employed, with a target depth of 67-70 nm. The sample was dry etched for 80 seconds, and a wet etch using 1:5 HF: H_2O for 5 seconds followed. For the source and drain vias, S1828 resist was spun, and the resist was exposed in the MLA with a 600 mJ/cm^2 dose, then developed for 55 seconds in 6:1 MF-319:H_2O. Etching of the second passivation layer followed the same steps as above, and exposed the first passivation layer below. The etching was then repeated for the removal of the first passivation layer, exposing the source/drain metal pads.

The vias were filled, and the field plates were deposited using LOR10B/S1813 lift-off. After resist coating, MLA exposure was done at 600 mJ/cm^2 dosage, and development was performed for 55 seconds in 6:1 MF-319:H_2O. A Ti/Pd/Au metal stack was deposited using the metal evaporator, with target thicknesses of 5/10/200 nm, respectively.

Chapter 4
Electrical characterisation

Electrical characterisation was performed using two probe stations: the TS2000-SE Semi-auto probe station at the Department of Electrical and Information Technology at LTH, and a Cascade Microtech Summit 11000B-M probe station with a Keithley 4200A-SCS Parameter Analyzer, at the Division of Solid State Physics at LTH. I-V sweeps measuring the output and transfer characteristics of the devices were performed, as well as breakdown voltage measurements of the devices and over TLM test structures. During device measurements, the transistor source terminal was grounded, and voltage biases V_G and V_D were applied to the gate and drain terminals, respectively.

A statistical analysis was performed with MATLAB on the measured data to analyse the effect of varying the gate recess depth, gate length L_g, source/drain recess depth, source-drain distance L_{sd}, field plate type, and device width. The following sections present and discuss the results. The methods used in extracting parameters from the raw data are covered in Appendix B. If not stated otherwise, the default device geometry between data points are as follows: a gate length of $L_g = 1$ µm, no field plates, 6 nm gate recess, 3 nm source/drain recess, source-drain spacing of $L_{sd} = 10$ µm, source-gate spacing $L_{sg} = 2.5$ µm, gate-drain spacing $L_{gd} = 6.5$ µm and device width of 100 µm.

4.1 Gate recess influence

The influence of gate recess variations on the transistor threshold voltage V_{th}, maximum transconductance $g_{m,\max}$, and on-current I_{on} was analysed and plotted in box plots, see Figure 4.1.

g_m and V_{th} values were extracted using transfer data gathered at $V_D = 3$ V. I_{on} was found at $V_G = 2$ V for the same $V_D = 3$ V transfer curves. A lack of data for the provided geometric restrictions required a wider range of source-drain spacings to be added to the analysis. No significant correlation between threshold voltage and varied source-drain spacing was observed, motivating this choice, see Figure 4.2. The plots in Figure 4.1 are thus composed of data from devices with varying L_{sd}.

There is a visible gap between V_{th} of the MIS and Schottky barrier (SB) devices, which appears to decrease with increasing recess depth. Furthermore, V_{th} trends towards less negative values with increasing recess depth for both device

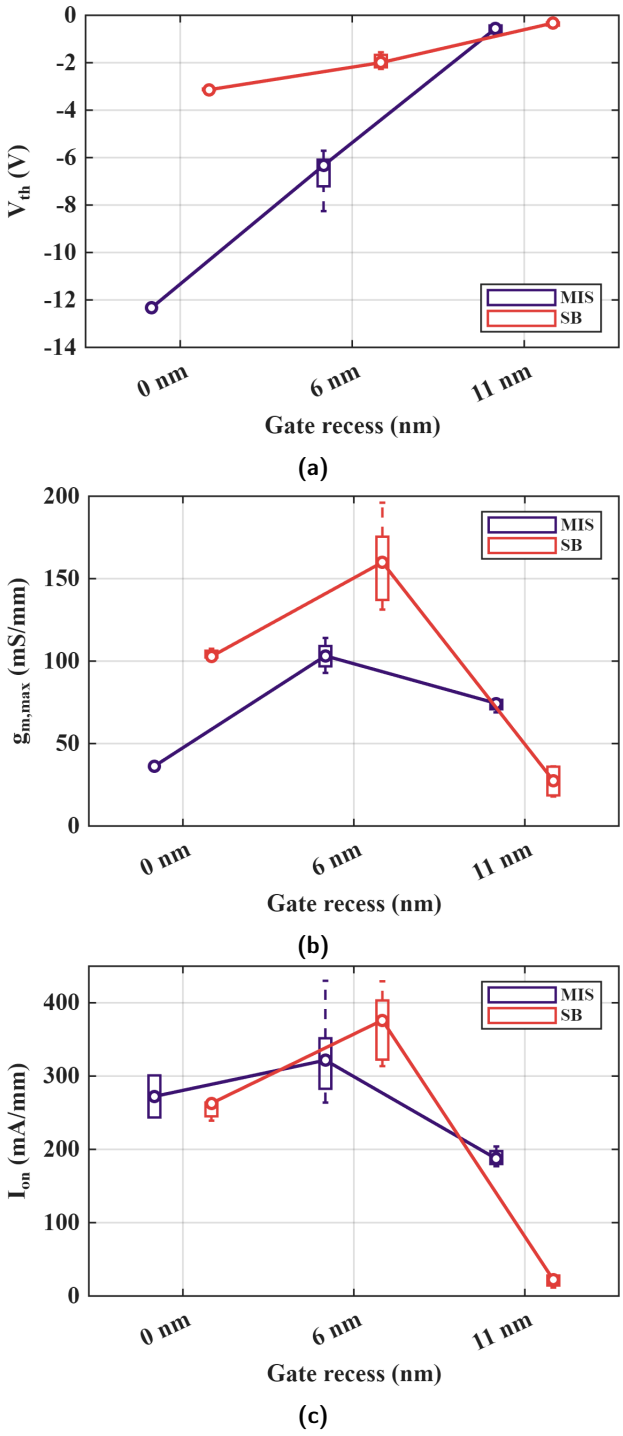


Figure 4.1: Three box plots presenting gate recess influence on a) the device threshold voltage V_{th}, b) the maximum transconductance $g_{m,\max}$, and c) the on-current I_{on}. The devices are grouped according to gate type.

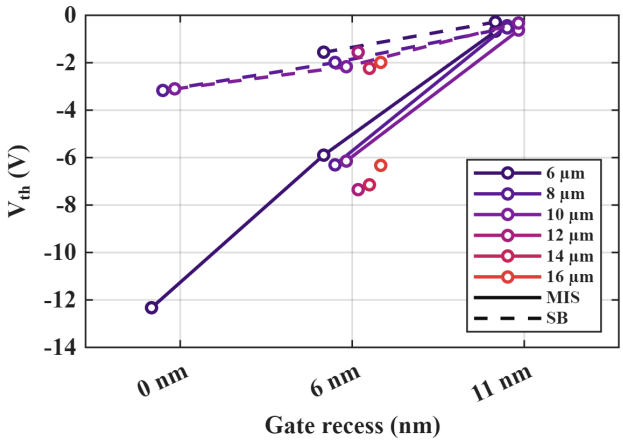


Figure 4.2: Plots showing the relation between median threshold voltage and gate recess at different source-drain spacings for MIS and SB devices.

types. This trend is consistent with the expected reduction in charge density from the AlGaN polarisation. The resulting decrease in 2DEG density requires a smaller negative gate bias to deplete the channel, moving the device towards normally off operation [39].

The $g_{m,\max}$ appears to exhibit a non-linear dependence on gate recess depth, with the highest median values observed for the 6 nm recessed devices compared to both the non-recessed and deeply recessed 11 nm devices. This behaviour suggests the existence of an optimal recess depth where gate electrostatic control is enhanced without significantly degrading the channel charge density. A shallow recess would reduce the AlGaN barrier thickness beneath the gate, thereby increasing the gate capacitance and strengthening the modulation of the 2DEG. An even deeper recess may begin to deplete the channel excessively, reducing the available carrier density and consequently lowering the achievable transconductance. Surface roughness induced by the gate recess etch could reduce the mobility of the carriers, contributing to the lowered transconductance. Between device types, the maximum transconductance is higher for the SB devices compared to the MIS devices for low-to-moderate gate recess, but the MIS devices provide more robust transconductance for the deep recess. Transconductance reductions and increases have both been attributed to recessed gates in the literature [40, 41].

The reduction in on-current with increasing gate recess depth further supports the hypothesised observation of the 2DEG depletion due to reduced polarization. A pronounced decrease in on-current is observed for the SB HEMTs at the 11 nm gate recess node. Together with the reduction in maximum transconductance, this suggests that the device is approaching the limits of normal transistor operation. The MIS devices similarly exhibit lowered on-currents at deep gate recess depths, although to a lesser extent than the SB devices. This may indicate that the observed current may be increasingly influenced by the MIS junction characteristics rather than transport through a highly conductive 2DEG channel.

The observed behaviour confirms that gate recessing is an effective method for threshold voltage engineering in AlGaN/GaN HEMTs. However, excessive gate recessing will lose the benefits of the high charge density 2DEG.

4.2 Gate length influence

The dependence of the threshold voltage V_{th}, maximum transconductance $g_{m,\max}$, subthreshold slope SS, and drain-induced barrier lowering DIBL on the gate length L_g was analysed and plotted in corresponding box plots, see Figure 4.3. A clear

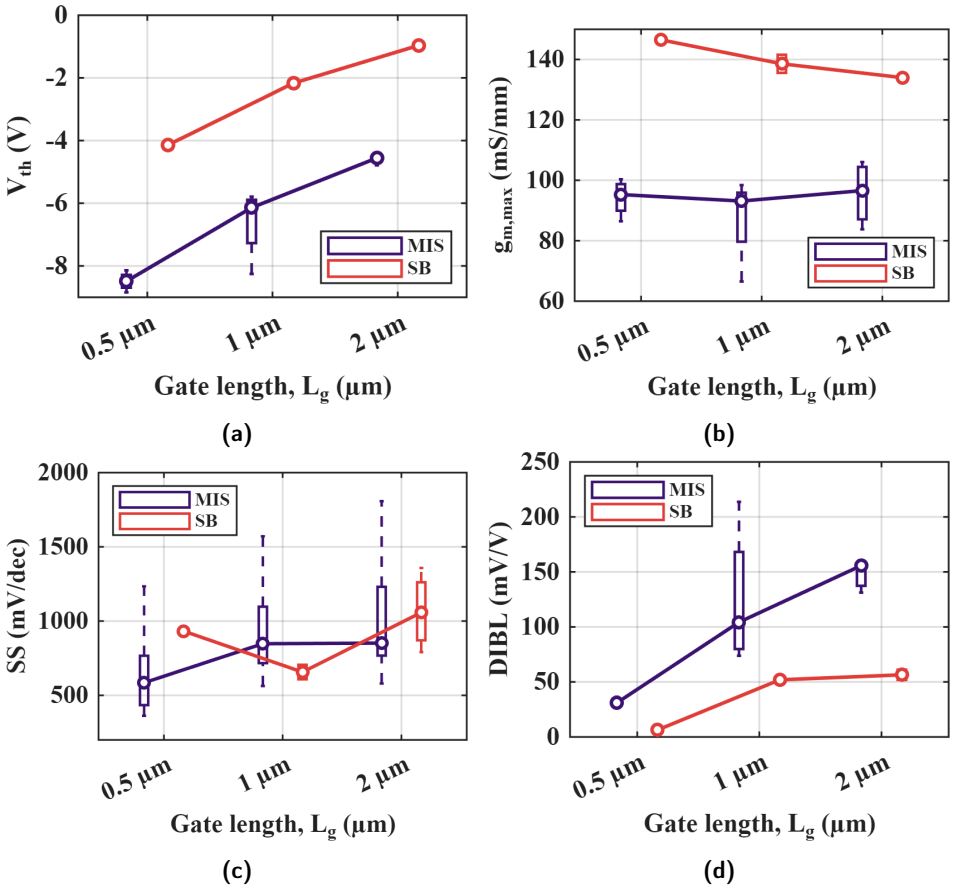


Figure 4.3: Four box plots showing gate length L_g influence on a) threshold voltage V_{th}, b) maximum transconductance $g_{m,\max}$, c) subthreshold slope SS, and d) drain-induced barrier lowering DIBL.

dependence of threshold voltage on gate length can be observed, with the median V_{th} in both MIS and SB type devices trending upward for increased L_g. This indicates that longer gate lengths provide stronger electrostatic control over the

channel and reduce the magnitude of the gate bias required to deplete the 2DEG. The V_{th} may be subjected to threshold roll-off, as observed in other HEMT structures [42, 43]. Furthermore, the threshold variations can be directly seen in the transfer characteristics for $V_D = 3$ V of the devices for different gate lengths, see Figures 4.4a and 4.4c. The corresponding output characteristics at $V_G = -3$ V highlight the difference in threshold voltage, with the MIS devices all conducting for the applied voltage, and only one of the SB devices doing the same, see Figures 4.4b and 4.4d.

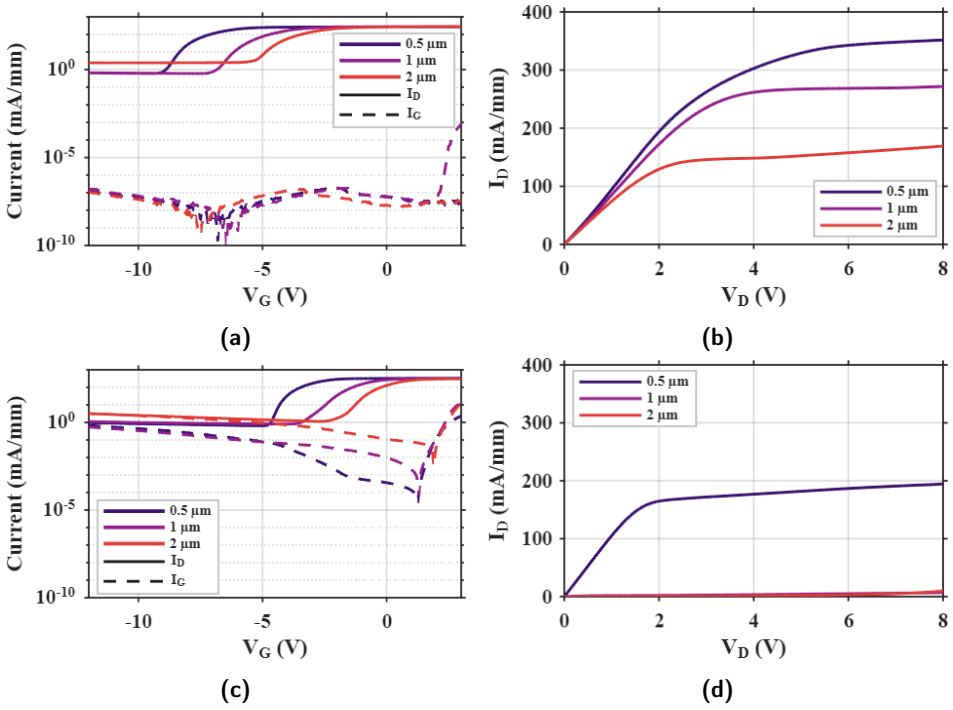


Figure 4.4: Curves displaying transfer characteristics for $V_D = 3$ V and output characteristics for $V_G = -3$ V for measured MIS and SB devices. a) Transfer characteristics for three MIS-HEMT devices with constant geometries except for the gate length being scaled. b) Output characteristics for the MIS-HEMTs. c) Transfer characteristics for three SB-HEMTs with varied gate lengths. d) Output characteristics for the SB-HEMTs.

A weak gate length dependence is observed for $g_{m,\text{max}}$ in the SB devices, with the median transconductance increasing as the gate length is reduced. This trend is expected, as shorter gate lengths generally provide improved gate control over the channel and reduced carrier transit times [11, 44]. However, the observed increase in transconductance does not appear to follow the simple MOSFET scaling relationship described by Equation 2.1, suggesting that additional effects such as access resistance may be influencing device performance.

In contrast, the $g_{m,\max}$ values of the MIS devices exhibit a weaker dependence on gate length. This suggests that the gate dielectric and associated interface properties may partially decouple the transconductance from geometric scaling, reducing the sensitivity of channel modulation to variations in L_g. Alternatively, the transconductance could be increasingly limited by factors other than gate control, such as the source/drain access resistances or significant gate leakage. The larger spread in observed values for the MIS devices could suggest the presence of process variations in the gate structures, possibly masking any added benefit in transconductance with gate length reductions.

The subthreshold slope appears to improve with decreasing gate length in both the SB and MIS devices, although a considerable statistical spread is present in the measured data. The extracted SS values remain substantially higher than the ideal limit of 60 mV/decade, indicating that the devices are not operating near state-of-the-art performance. This suggests that non-idealities such as interface traps or gate leakage influence the subthreshold performance [45].

Similarly, the DIBL is observed to decrease with reduced gate length. This trend is contrary to the behaviour typically expected in short-channel devices, where DIBL generally increases as the gate length is scaled down due to enhanced drain control of the channel potential [11, 46]. One possible explanation is that the investigated gate lengths remain sufficiently large that short-channel effects have not yet become the dominant factor governing device operation. In this regime, improvements in gate control associated with reduced gate length may outweigh any increase in drain-induced electrostatic coupling.

4.3 Source/drain recess influence

The on-resistance of the device was plotted against the source/drain recess depth, see Figure 4.5. The on-resistance R_{on} was captured from output characteristic curves corresponding to $V_G = -2.625$ V for devices with geometries $L_g = 0.5$ µm, $L_{sd} = 12$, 14, and 16 µm. The on-resistance decreases with recess, which is the expected result [27]. However, little usable data could be collected from the 5 nm devices, as these devices were either susceptible to breaking or were short-circuited due to processing errors. The data can therefore not be considered conclusive, and further studies and optimisations of the source/drain recess are a natural continuation of this project.

The difference in R_{on} magnitude between the MIS and SB devices can be explained by the method of analysis, with the comparison utilising a constant V_G instead of $V_G - V_{th}$. At $V_G = -2.625$ V, the MIS device and SB device will not be turned on to the same extent, as they have different threshold voltages. Furthermore, the on-resistance for the MIS device is larger than seen in the TLM measurements performed before the added gate stack and vias, refer to Figure 3.1. One explanation is that the vias added significant resistance to the contacts, increasing the effective on-resistance of the HEMTs. Another reason could be the gate recess. The analysed devices have a gate recess of 6 nm. The consequent reduction in 2DEG sheet density could thus explain the reduction in on-resistance.

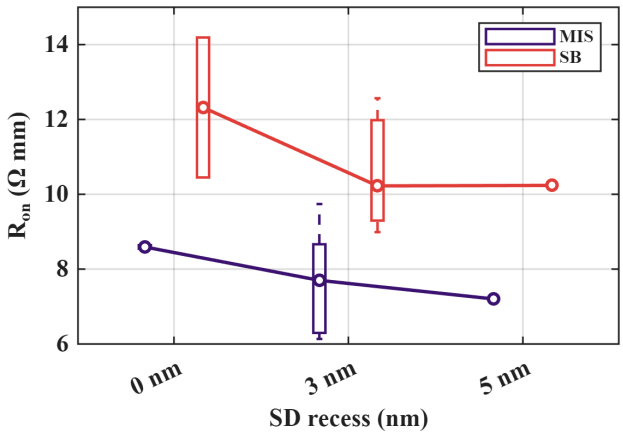


Figure 4.5: Box plots showing the influence of source/drain recess on on-resistance R_{on} for MIS gate devices and Schottky-barrier gate devices.

4.4 Source-drain length influence

The influence of the source-drain length L_{sd} on the subthreshold slope and R_{on} was analysed and plotted in box diagrams, see Figure 4.6.

The median subthreshold slope is observed to decrease with shorter L_{sd}, except for a smaller value at $L_{sd} = 16$ µm and a slight increase at $L_{sd} = 6$ µm. The on-resistance is observed to increase significantly going from an $L_{sd} = 12$ µm to 10 µm. This discrete jump in resistance could be explained by the lack of devices measured at lower L_{sd} nodes, perhaps skewing the result. Probe resistances could also be a contributor, as measurements for the lower source-drain spacing nodes were performed with a different probe station than the other devices. Disregarding the discrete jump, the on-resistance tends to decrease with reduced spacing. Shorter source-drain distances reduce the length of the access regions between the channel and the ohmic contacts when the gate length is kept constant, thereby reducing the resistances that mask the channel [47, 48]. This could further explain the improvement in SS with reduced source-drain spacing. Less access resistance could lead to improved gate control over the channel current, and thus a higher SS, as more gate voltage falls directly over the channel.

4.5 Device width influence

The two device widths of 100 and 200 µm were compared using on-current I_{on} at $V_D = 3$ V and $V_G = 2$ V, maximum transconductance $g_m, \max$ for $V_D = 3$ V, and output conductance g_d for $V_G = -2.625$ V, see Figure 4.7. The device on-current and the maximum transconductance, normalised with the device width, scale almost proportionally with the width. The median current and transconductance for the MIS and SB devices remain approximately constant between the 100 µm and 200 µm variations. This result is consistent with the expected scaling

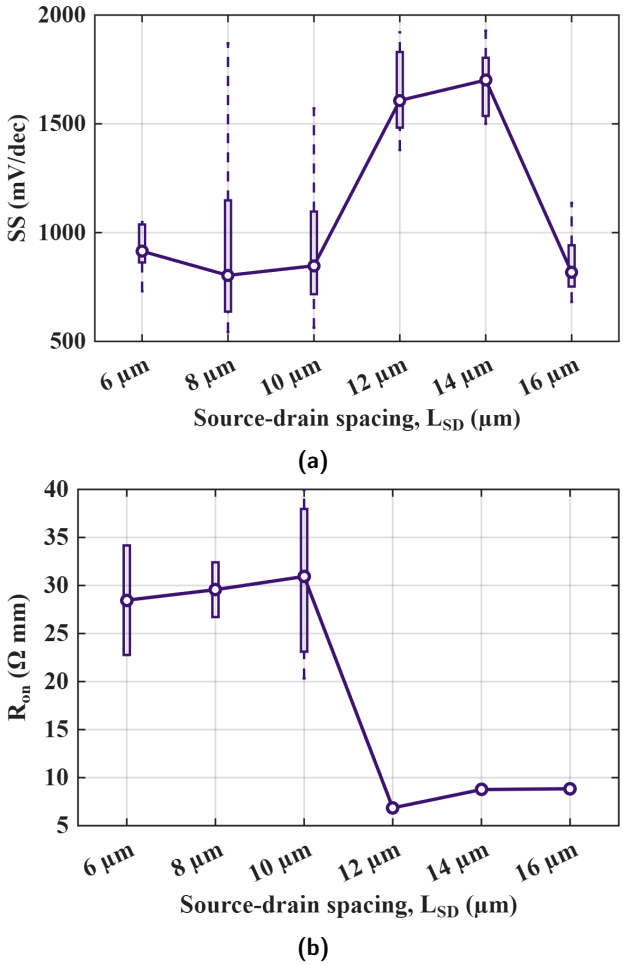


Figure 4.6: Box plots showing the influence of source-drain spacing on the MIS devices' a) subthreshold slope magnitudes and b) on-resistances R_{on}.

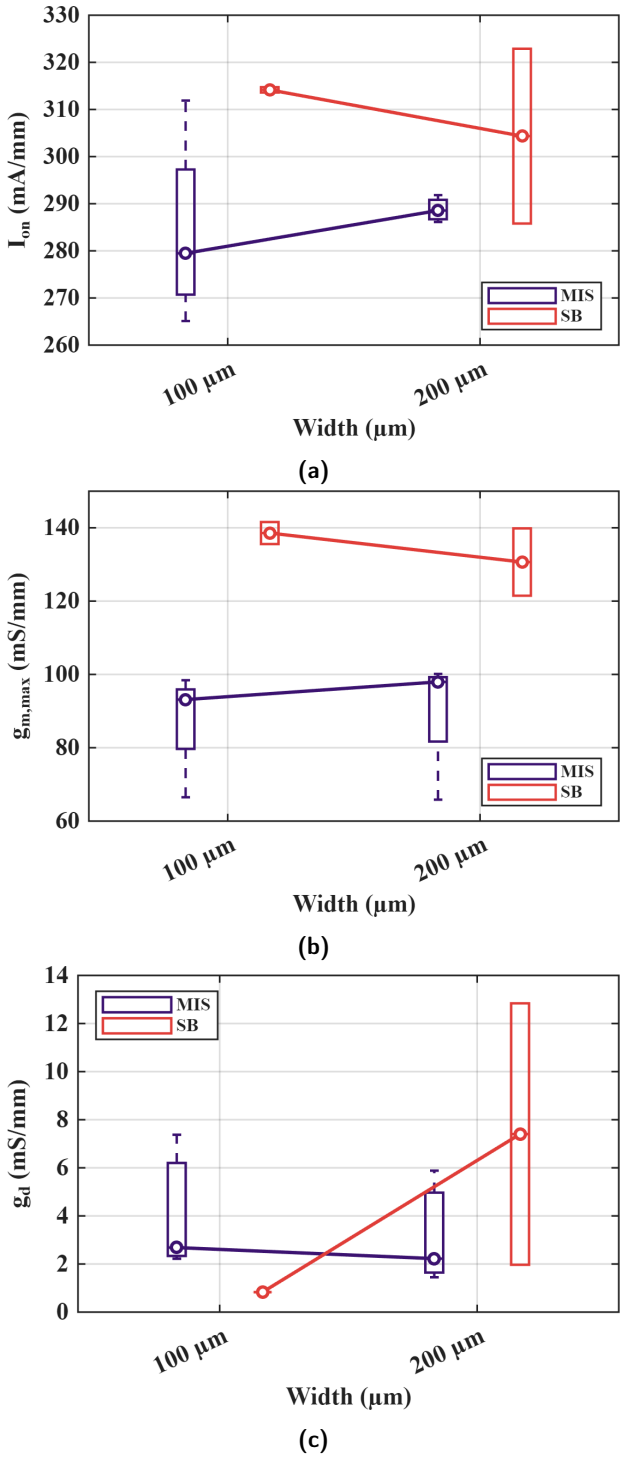


Figure 4.7: Box plots showing device width W influence on a) on-current I_{on}, b) transconductance g_m and c) output conductance g_d.

behaviour of HEMTs and general FETs, as the total channel current capability is proportional to the effective gate width, see Equation 2.1. Since transconductance is fundamentally related to the rate of change of drain current with gate voltage, a doubling of channel width ideally results in a corresponding doubling of g_m.

Regarding the output conductance, the median and the statistical spread stay constant for the MIS devices, but there are deviations between the 100 µm and 200 µm SB devices. One cause could be that potential process variations, influencing the drain terminal's control over the device, assist in blurring an otherwise linear scaling relationship in output conductance.

In general, the observed scaling indicates that the channel conductivity and gate control remain relatively uniform across different device widths. The observed on-currents are furthermore, for both SB and MIS devices, in the expected range for moderate-performing AlGaN/GaN HEMTs compared to the literature [49–52].

4.6 Field plate influence

The subthreshold slope, output conductance g_d, DIBL, and off-current I_{off} were analysed between the source field plate (SFP), gate field plate (GFP), gate-drain field plate (GDFP), and no field plate (NoFP) MIS-HEMT device designs. The off-current was found for $V_G = -12$ V in the transfer characteristic data gathered at $V_D = 3$ V. The results are seen in Figure 4.8. The subthreshold slope is highest for the SFP and GDFP devices, while the lowest median value is observed for the devices without field plates. The large spread observed for the NoFP devices suggests the presence of additional sources of variability, or that the presence of field plates could stabilise gate control at the cost of SS performance. A leaky or non-uniform SiN_x passivation layer could degrade gate control and contribute to the observed spread in subthreshold characteristics.

The SFP devices exhibit the highest median output conductance, whereas the remaining field plate configurations show lower output conductance and considerably less variation. The NoFP devices again exhibit high spread, highlighting their variability. Increased output conductance of the SFP devices implies a reduced gate control of the channel and a stronger influence of the applied drain voltage. It should be noted that a limited amount of data could be gathered regarding the field plate devices' values, and the high SFP output conductance could be the product of statistical spread.

The DIBL results support this interpretation. The lowest median DIBL is observed for the GDFP devices, while the GFP and no field plate devices exhibit comparable performance. In contrast, the SFP devices show substantially larger DIBL values. As the GFP and GDFP devices do not suffer from the same short-channel effects, the presence of the source-connected field plate extending above the gate region, or process variations associated with its fabrication, may be responsible for the observed deterioration in SFP performance.

The off-state current measurements show a similar trend to the SS. The lowest median off-current is observed for devices without field plates, while all field plate designs exhibit increased leakage. Together with the elevated SS, output conductance, and DIBL, this observation further indicates that the SFP devices

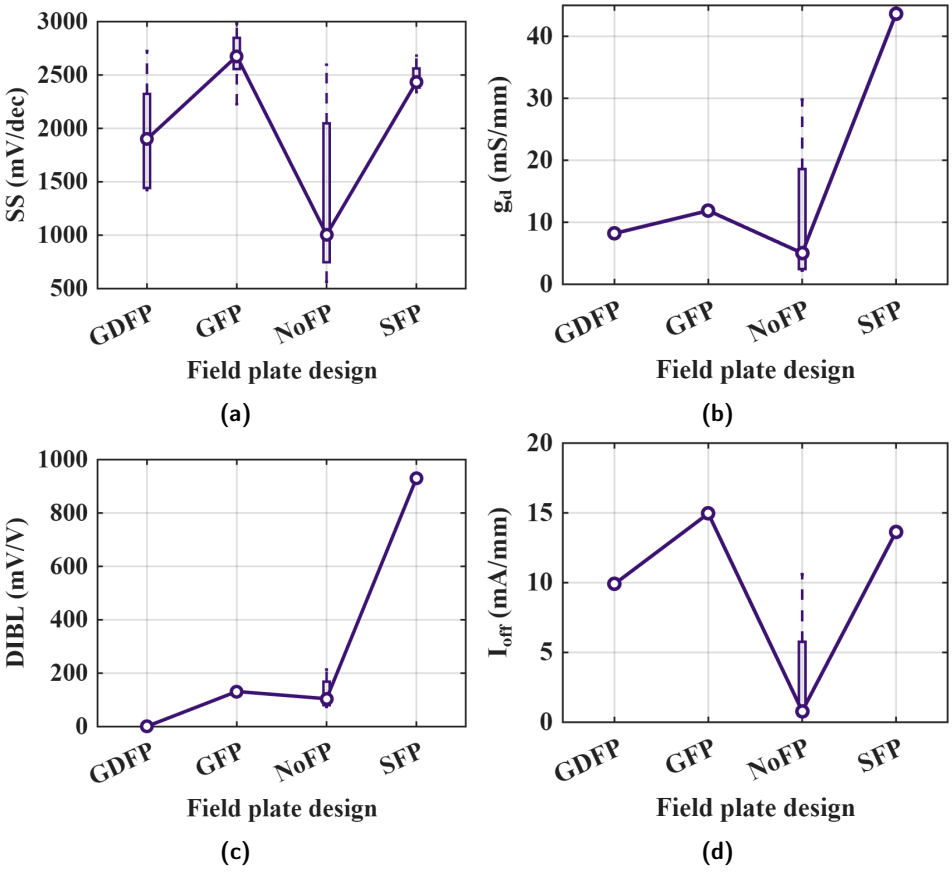


Figure 4.8: Four box plots showing the field plate influence on a) subthreshold slope, b) output conductance g_d, c) drain-induced barrier lowering DIBL, and d) off-current I_{off}.

experience reduced gate control and increased susceptibility to leakage mechanisms compared with the other field plate configurations.

4.7 HEMT and MESA breakdown measurements

Off-state breakdown measurements from 0 V to 100 V were performed for $V_G = -12$ V on the different field plate design variations, see Figure 4.9. The observed

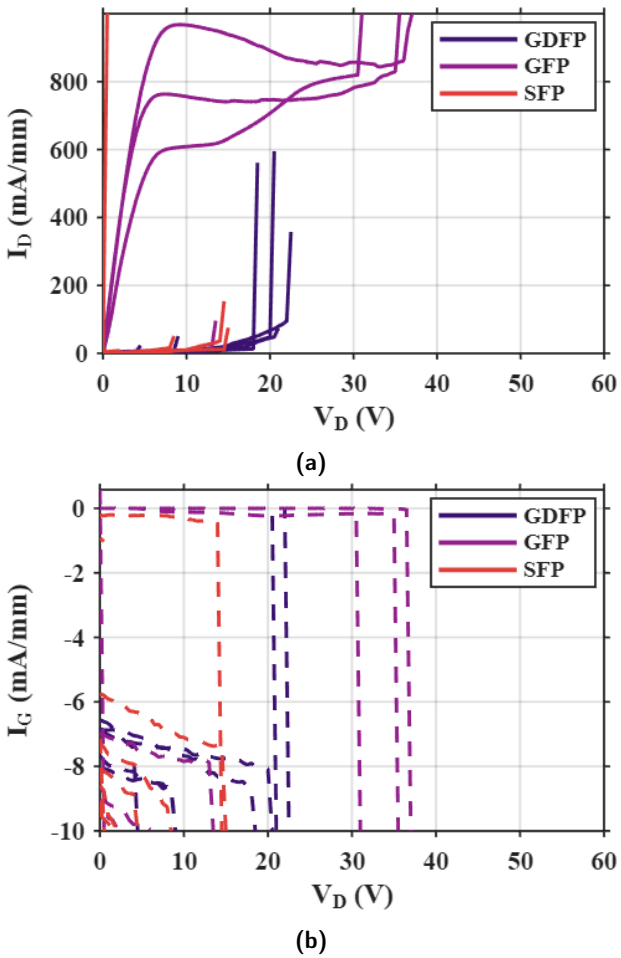


Figure 4.9: Breakdown sweeps of MIS-HEMT devices with gate-drain field-plates (purple), gate field-plates (magenta), source field-plates (red), displaying a) the drain current and b) the gate current versus voltage.

data show rapid discrete increases in drain and gate currents with voltage. The increase in current stops when the measured value reaches a user-defined maximum, or compliance. For the drain current, the absolute value of the compliance was set

to 1000 mA/mm, and for the gate current, it was set to 10 mA/mm. When one of the two currents reaches compliance, the device is counted as broken.

In Figure 4.9a, most SFP and GDFP devices never reach drain compliance, as the gate leakage seen in Figure 4.9b hits compliance before the drain current, resulting in a device breakdown through the gate. The GFP design appears to be the most robust, as the GFP devices are observed to suppress heightened gate currents and high drain currents for the highest voltage range.

Collectively, these off-state breakdown voltages are all lower, many significantly lower, than 40 V. This is far from the expected performance reported in the literature [53–55], with a possible reason being the low gate-drain spacing compared to conventional lateral devices [56]. Another possible contributor is the high gate leakage, as many measured MIS devices demonstrated high gate leakage, despite the added insulator in the gate stack. To illustrate this, measured gate currents from the transfer characteristics for leaky MIS devices can be seen in Figure 4.10, corresponding to the different field-plate designs. The gate leakage is

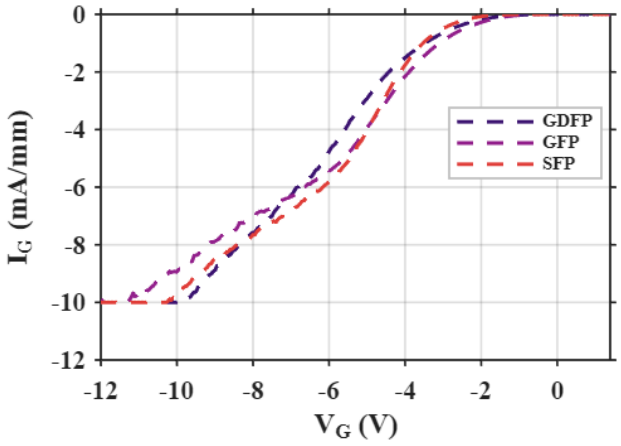
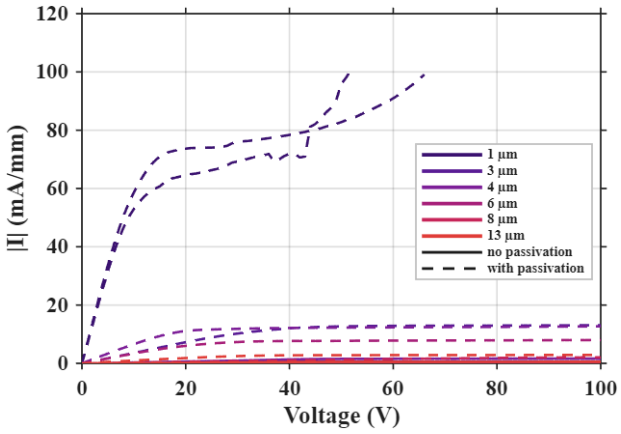


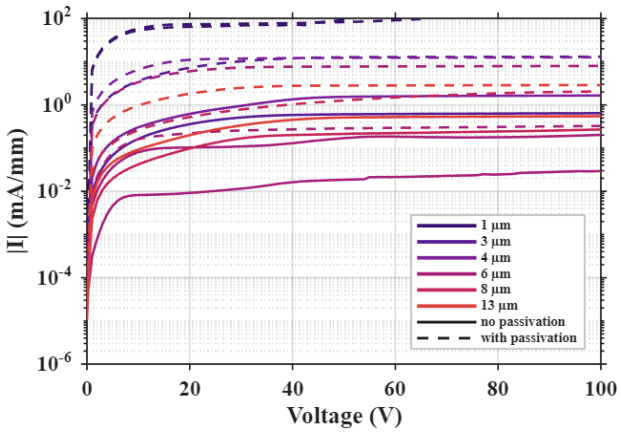
Figure 4.10: Transfer gate current characteristics of a MIS-HEMT showcasing leaky behaviour, with high gate currents present when the device is turned off. In the showcased device, the SFP and GDFP devices hit compliance after a gate voltage of ≈ -8 V is applied, and the GFP device is close to compliance at -12 V.

theorised to be caused by the LT-GaN buffer layer situated between the UID-GaN channel and the substrate. A leaky buffer layer could introduce a leakage path between the gate contact pad and the source terminal.

To search for further reasons for the early breakdown of the devices, two sets of TLM test structures were measured. MESA breakdown tests were conducted between pads separated by 1 µm deep MESA isolation trenches of different widths. One set had the SiN_x passivation layer on top, and the other had no passivation. See Figure 4.11 for the results of the measurements. The breakdown results revealed that the SiN_x passivation layer appears to increase the devices' susceptibility to breaking. The 1 µm isolation trenches are seen to hit the compliance



(a) Linear scale



(b) Log scale

Figure 4.11: Breakdown sweeps of MESA structures separated by MESA isolations of varying width, one set with a SiN_x passivation layer present and the other without the passivation layer. The data is presented twice, as a) a linear plot and b) a log-scale plot.

of 100 mA/mm at 40 to 60 V when a SiN_x passivation is present. Without the passivation, no isolations are broken. Furthermore, the passivated trenches appear to allow higher current magnitudes to pass between the MESA pads than for the unpassivated trenches. The gate pads of the devices are isolated from the source and drain pads by 1 µm wide MESA rings and further separated by the SiN_x passivation. The high leakage observed in these MESA measurements may point to the cause of high gate currents between the gate and source/drain contact pads, causing early breakdown and non-ideal device performance.

Chapter 5
Conclusion

The AlGaN/GaN HEMT fabrication process successfully produced functioning transistor devices exhibiting typical on-current and transconductance performance. Increased gate recess depth was observed to shift the devices towards normally-off behaviour, while deeper source and drain recesses reduced the device on-resistance. These trends are consistent with previous literature, indicating that future process development should investigate deeper source and drain recesses together with optimized contact annealing recipes to further reduce contact resistance.

The unexpected high conductivity of the UID-GaN channel layer led to significant leakage currents, and is theorised to be a contributor to the high 2DEG electron sheet density of the provided material stack. Mitigation of this issue through deeper MESA isolation etching introduced additional complexity to the gate fabrication process. These results suggest that future devices fabricated with similar material stacks would benefit from a thinner UID-GaN channel layer, enabling complete isolation etching while maintaining simpler, more reliable lithography processes.

High gate leakage currents and early device breakdown were observed despite efforts to increase device isolation using ion implantation and deep MESA isolation rings. This was attributed to a leaky LT-GaN buffer layer as well as an insufficient SiN_x passivation quality. The SiN_x deposition optimisation treated the deposition of SiN_x on Si samples, likely not reflecting the same conditions as imposed by a deposition of SiN_x on GaN. As a logical next step of this thesis, SiN_x deposition optimisation using PECVD on GaN substrates should be performed.

In addition, greater robustness in device layout design is required to more effectively study scaling effects. Variations in spacing and gate dimensions between fabricated devices limited the ability to isolate individual scaling parameters.

Overall, this work demonstrates the successful fabrication and characterisation of AlGaN/GaN HEMT devices while identifying several critical process limitations affecting leakage currents, breakdown behaviour, and device variability. The findings can guide future process optimisation and contribute to the continued development of reliable and scalable GaN-based transistor technologies.

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Appendix A

Metal contact salvage trials

A.1 Annealed metal stripping

To remove the non-ideal electrical contacts, an attempt to remove the deposited metal pads using wet etching methods was undertaken. To avoid possible damage to the main sample, a sample of AlYN/GaN and one of AlScN/GaN, both with similar contacts of Ti/Al/Ni/Au, were used as tests for the metal stripping. Initial trials on one of the samples used Au etchants, Ni etchants, Al etchants, and Ti etchants, separately. After these trials, no visible change had taken place, concluding that annealing had formed alloys that required other etchants to remove.

Aqua regia solution was formed using 3:1 HCL:HNO_3, and the sample was submerged in the etchant for 10 minutes. Optical microscopy revealed a visible difference before and after the aqua regia treatment. The sample was then subjected to an additional 10-minute etch, with more visible change after. A final 10-minute etch was done, which showed less change. The entire stack was not theorised to have been completely removed; possible Ti/TiN remained on the pad regions, and a stronger Ti etchant than in the previous etching attempt was used, being a 1:2 HF:HNO_3 solution. The sample was submerged for 10 minutes, and it was concluded that the solution was capable of removing the remaining metal alloy.

A second test sample was used to verify and tune the striping method. A cyclic approach was used, with the sequence of submerging the sample in aqua regia for 30 minutes, changing to a 30-minute submersion in 1:2 HF to HNO_3, and repeating once.

The two metal-stripped test samples were subsequently characterised in a Hitachi SEM SU8700, where EDS analysis indicated the presence of Ga rich regions at the locations of the stripped pads. As the pads were located in an AlYN or AlScN layer, respectively, the appearance of Ga was likely due to material diffusion during the annealing step. On the AlScN sample, the Ga was removed by submersion in heated TMAOH at 60°C for one minute. Analysis using a profilometer showed that the removal of Ga led to deep holes into the substrate, on the order of 100 nm. It was theorized that a complete recess of the pad regions could help in the formation of good sidewall contacts to the metal-stripped HEMT samples.

To perform a recess on the stripped samples, the same recipe as the one developed for the AlGaN recess, seen in section 3.1.1, was used. Three test samples

were used: the already-stripped AlYN sample, the stripped and TMAOH-treated AlScN sample, and a non-TMAOH-treated AlScN sample that was stripped cyclically with 10-minute submersion times instead of 30 minutes. These samples were coated with S1813 resist, patterned by photolithography and MF-319 development, and subjected to different etching times. The AlYN sample was etched for 2 minutes, but resist removal and profilometer analysis revealed no visible etching. The AlScN samples were etched in parallel using the same method. The TMAOH-treated AlScN sample was the only one that showed evidence of etching.

A second recess etch trial was conducted on a TMAOH-treated AlYN sample. An etch for 5 minutes was performed, and after resist removal and profiling, the etching was found to have formed recesses of an average of 20 nm in depth. It was thus assumed that TMAOH treatment was effective in completely removing the alloyed region. However, further trials suggested that TMAOH could assist in the complete removal of the AlYN or AlScN barrier layer.

A.2 Contact redeposition

Three samples were to be metal stripped, treated with TMAOH, recessed, and have new source-drain pads deposited, to analyse if metal stripping and redeposition would be feasible for the main AlGaN sample. Two AlScN samples, one being the previously TMAOH-treated sample and the other being a newly stripped sample, and one TMAOH-treated AlYN sample, were used in the redeposition trials. After the S1813 resist coating, photolithography, and development, the samples were to be etched. The previously used AlScN sample was etched for 5 minutes and 30 seconds to fully recess the barrier layer and facilitate sidewall contact formation, while the two remaining samples were etched for 4 minutes.

After etching, the sample was cleaned, and a lift-off resist coating of LOR10B and S1813 was spun on, patterned with photolithography, and developed in 6:1 MF-319 to H_2O development solution. The samples were then placed in an evaporator to deposit a Ti/Al/Ni/Au stack of 30/100/40/40nm thickness, respectively. After lift-off, electrical characterisation using a four-point probe measurement of the resistance of the TLM structures was performed. Each of the samples showed poor performance, with resistances on the order of 10 kΩ, significantly higher than the acceptable sub 100 Ω resistances. An annealing step was performed at 750°C for one of the AlScN samples and the AlYN sample, while the remaining AlScN sample was subjected to a lower temperature, 500°C anneal. Electrical characterisation after the annealing indicated that the performance was still not ideal, with kΩ resistances present. It was thus concluded that metal stripping and redeposition were not to be performed on the AlGaN sample.

Appendix B

Electrical parameter extraction

The following chapter exemplifies how the electrical data presented in Chapter 4 of this text was extracted. Data extraction was done using MATLAB.

B.1 Threshold voltage and transconductance

The threshold voltage and transconductance were extracted using the transfer characteristics, as described by [57]. $g_m, \max$ was found as the inflection point of the transfer curve. A tangent line was fitted to this inflection point, and its intersection with the x-axis returned the estimated value for V_{th}, see Figure B.1.

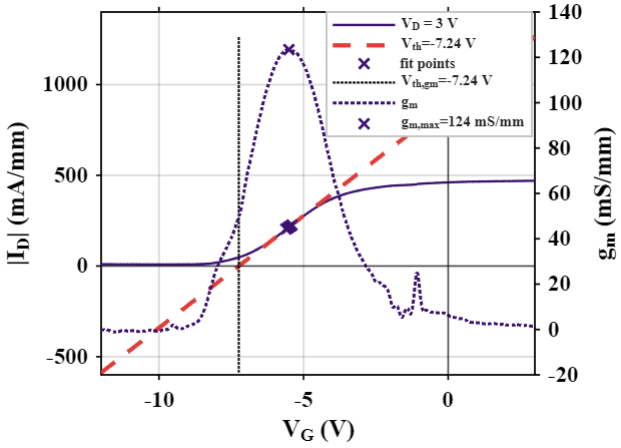


Figure B.1: Example V_{th} and g_m extraction from measured transfer data of one of the MIS-HEMT devices for $V_d = 3$ V.

B.2 Subthreshold slope and DIBL

The extraction of the subthreshold slope was performed by finding a linear window in a logarithm-scaled transfer plot, $\log I_D$ to V_G, fitting points to the windowed

line, and calculating the average slope between the points. A plot of an SS extraction, visualised with points in the detected linear window, is seen in Figure B.2.

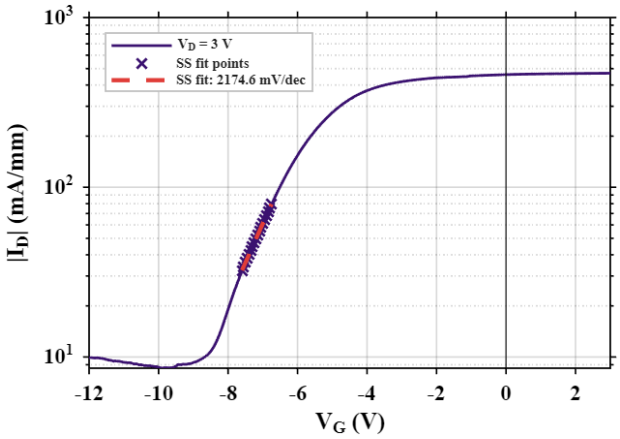


Figure B.2: Subthreshold slope extraction plot for transfer data at $V_D = 3$ V from a measured MIS-HEMT device.

DIBL was calculated using the V_{th} extraction mentioned above for two values of V_D, picked to be 1 and 5 V. The DIBL is found as the slope of the line fitted between the V_{th} data points, see Figure B.3.

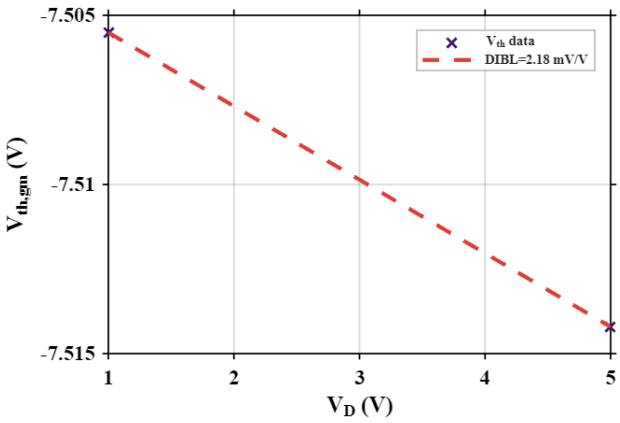


Figure B.3: Extraction of DIBL from electrical data. The threshold voltage values were extarcted using the g_m inflection point method for $V_D = 1$ V and $V_D = 5$ V.

B.3 On-resistance and output conductance

The on-resistance and output conductance were extracted from the output characteristics of the transistor data: the on-resistance as the inverse slope of the low-V_D linear regime, and the output conductance as the slope of the high-V_D saturation regime, see Figure B.4.

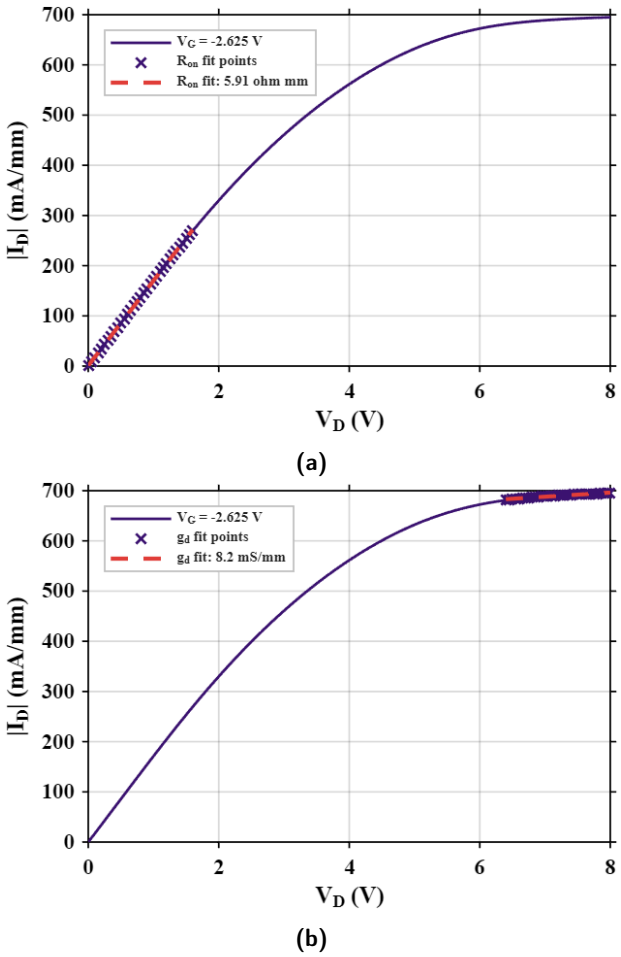


Figure B.4: Plots showing extraction of a) the on-resistance, and b) the output conductance, from electrical HEMT output characteristics for $V_G = -2.625$ V.

B.4 On- and off-currents

The on- and off-currents were extracted from the transistor transfer characteristics, with the on-current being found at the highest $V_G = 2$ V and the off-current at

the lowest $V_G = -12$ V, see Figure B.5.

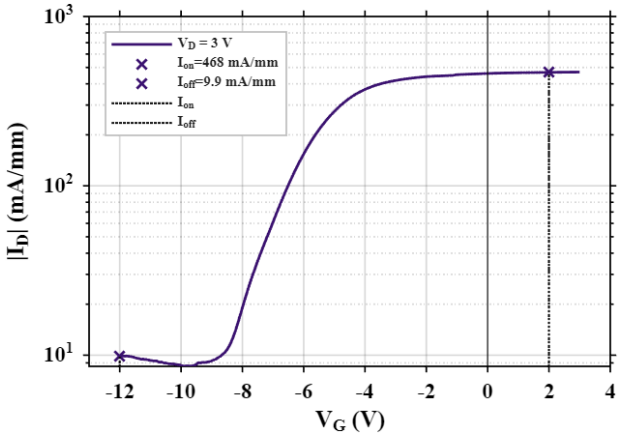


Figure B.5: Plot indicating the extraction of on- and off-currents from HEMT transfer characteristics.